



A review of digital twins in smart industries: Concepts, milestones, trends, applications, opportunities and challenges

Peipei Ding^a, Shi Qiang Liu^{a,*}, Raymond Chiong^{b,c,d,**}, Sandeep Dhakal^{b,d}, Dewang Chen^e,
Debiao Li^a, Hoi-Lam Ma^f, Sai-Ho Chung^g

^a School of Economics and Management, Fuzhou University, China

^b School of Science and Technology, The University of New England, Australia

^c School of Medicine and Public Health, The University of Newcastle, Australia

^d College of Management, Sichuan Agricultural University, China

^e School of Transportation, Fujian University of Technology, China

^f Department of Supply Chain and Information Management, The Hang Seng University of Hong Kong, Hong Kong

^g Department of Industrial and Systems Engineering, Hong Kong Polytechnic University, Hong Kong

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ABSTRACT

A digital twin (DT) is a real-time, highly accurate, virtual replica that reflects the states and behaviours of physical objects or systems. DTs can enable monitoring, simulation, prediction, optimisation as well as the structured integration of technologies, data flows and functional processes within smart industries. In recent years, the DT technology has emerged as a research hotspot, which has prompted us to conduct a review of its development and application in various industries. We have identified 30 leading journals that have significantly contributed to DT research, with the *Computers in Industry* (CII) journal ranking second among these 30 journals with more than 80 related publications. After briefly discussing the key concepts and major milestones around the development and rapid adoption of DTs in smart industries, we focus on reviewing and analysing the DT publications from the CII journal from 2018 to present by systematically categorising them into four primary application domains: manufacturing, construction, transportation, and technologies and paradigms. We also discuss potential research opportunities (e.g., life cycle management, cross-disciplinary integration, human-machine collaboration) and challenges from a theoretical perspective, and provide managerial insights (e.g., building open standards, enhancing data access compatibility, extending DTs' operational functions, applications to more industries) from a practical perspective. This review will be helpful for academic researchers and industrial practitioners to gain a broad understanding of the versatility of DTs, thereby fostering interdisciplinary innovation.

1. Introduction

A digital twin (DT) is a highly realistic and dynamic digital representation of a physical entity featuring real-time two-way communication between the physical entity and its virtual counterpart. With the rapid advancement of multidisciplinary technologies and the increasing need for high-precision management, the research into and development of DTs have been gaining popularity in both industry and academia. The DT paradigm facilitates the intelligent transformation of various industries by connecting the physical and virtual worlds through digital technologies, big data, cloud computing, the Internet of Things (IoT),

modelling, simulation, optimisation and artificial intelligence (AI). Currently, DT applications have achieved significant success in various sectors, including manufacturing (Cimino et al., 2019; Perno et al., 2023), construction (Long et al., 2024), agriculture (Mitsanis et al., 2024), finance (Chatterjee et al., 2024), education (Singh et al., 2022), smart cities (Bibri et al., 2024), healthcare (Rowan, 2024) and mining (Plavšić and Mišković, 2024).

The DT technology is rapidly transforming various industries (Kober et al., 2024). With the rapid expansion of DT research, systematically analysing its academic landscape is crucial to drive further advancements in the field. This can help identify the key issues, emerging trends

* Corresponding author.

** Corresponding author at: School of Science and Technology, The University of New England, Australia.

E-mail addresses: samsqliu@fzu.edu.cn (S.Q. Liu), rchiong@une.edu.au (R. Chiong).

and unexplored areas; provide strategic guidance for future research; and eventually foster collaborative innovation between academia and industry. Accordingly, our study addresses two primary research questions. First, what are the foundational concepts, key processes and milestones that have shaped the development of DTs? Second, what are the current trends and research potential in DT research? We have identified influential journals and selected *Computers in Industry* (CII) as a representative leading journal for a detailed literature review. This analysis evaluates the current state of DT research, explores emerging opportunities and challenges and provides managerial insights to guide future academic inquiries and practical applications.

1.1. Definition of digital twin

The goal of a DT is to create a precise, dynamic and real-time virtual replica of a physical entity, process, system or environment by simulating behaviours, characteristics and performance of a physical entity under various scenarios (Tao et al., 2022). The physical entity could be anything from a product or a process to a system or even a natural environment. Originally, the Defence Acquisition University defined DT as “an integrated multi-physics, multi-scale, probabilistic simulation of an as-built system, enabled by Digital Thread that uses the best available models, sensor information and input data to mirror and predict activities/performance over the life of its corresponding physical twin” (Cameron et al., 2018). It should be noted that, as the use of DTs has expanded beyond the original industrial and manufacturing sectors, the definition and scope of a DT has also become more flexible. For example, Verdouw et al. (2021) defined an “Imaginary DT”, where the physical counterpart of the digital entity does not yet exist (such as designed, but not yet an existing object).

As illustrated in Fig. 1, a typical DT system comprises three main components: the physical space, the corresponding virtual space, and a communication channel that facilitates the exchange of information between the physical entities and their virtual counterparts. The physical space includes real-world entities such as equipment, environments and humans, which generate real-time data through sensors and IoT devices. The virtual space employs highly accurate mathematical models to simulate the behaviours and performance of these entities under various conditions. The communication channel establishes a two-way flow of data and information between the physical entities and

their virtual counterparts. In this regard, Kritzinger et al. (2018) differentiate DTs from “digital models” and “digital shadows” based on the level of information flow between the physical and virtual counterparts. A digital model lacks any automated exchange of information between the physical and virtual counterparts; a digital shadow features automated information flow from the physical to virtual entities; and a DT features full bi-directional flow of information.

In a DT system, metrological methods create accurate virtual counterparts of physical entities, enabling physical-to-virtual twinning. These virtual models, in turn, optimise and control physical systems, achieving virtual-to-physical twinning. This bidirectional interaction between physical and virtual spaces exemplifies the DT concept, and the real-time data flows enable synchronisation and control of physical objects through their virtual counterparts (Kritzinger et al., 2018). Big data from the physical domain is continuously fed into the virtual entity, constructing a virtual counterpart that provides decision support for optimising physical operations. The information exchange between physical and virtual counterparts is inherently bi-directional. Real-time data from the physical system continuously updates the virtual model, ensuring an accurate representation. In return, insights, simulations and predictions generated by the virtual model feed back into the physical system, guiding its control and optimisation. The frequency and volume of data transfer depend on the specific application and system. For instance, industries like automotive or aerospace may collect and transmit real-time data every few milliseconds. In contrast, systems such as city infrastructure or climate models may only require data updates at longer intervals (hourly or daily). Within the closed-loop optimisation system, continuous data supply from the physical space ensures high-fidelity mapping in the virtual model. The twins evolve dynamically, refining their accuracy as they constantly interact with the physical world.

The accuracy of a DT depends on several factors, which include the quality of input data, the sophistication of the models and the computational power. Some DT models are simplified and primarily used for strategic planning, lacking the ability to capture every nuance of the physical system. In contrast, other DT models used in fields like aerospace or nuclear power are highly detailed and precise due to the critical need for accuracy. While continuous data updates improve fidelity over time, uncertainties and incomplete data can still limit their precision.

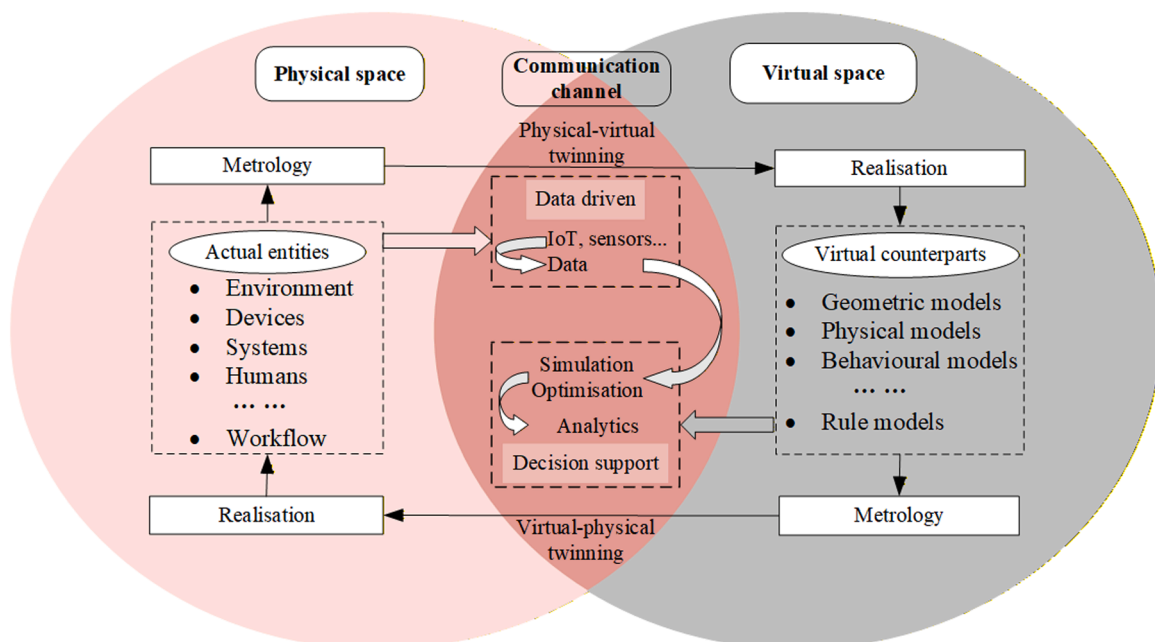


Fig. 1. Main components of a typical DT system.

1.2. Workflow of a digital twin

The workflow of a typical DT system is depicted in Fig. 2. We can see that a substantial amount of data is obtained from various sources for establishing the digital-driven DT and its associated models. As with most models, the data must be cleaned and pre-processed to eliminate deviations and outliers before being integrated with the DT. Once a complex DT model is established, a consistency test must be conducted to eliminate potential conflicts and ensure that the DT aligns with the actual requirements. Following the completion of a consistency test, the DT leverages data analysis to simulate physical entities and optimise their operational efficiency. This approach enables precise control of equipment and resources across various stages within the physical environment, facilitating informed decision-making in areas such as predictive maintenance, demand forecasting, safety management and operational adjustments. The DT serves as a virtual representation of the physical system for real-time, data-driven monitoring, optimisation and simulation throughout its entire lifecycle.

Prior to the construction of a DT system, it is necessary to clearly define each physical entity that needs to be digitally mirrored. The complex physical entities must provide rich data resources that are easy to collect and synchronise into the digital entity in real time. The physical entities also need to be optimisable, have significant enhancement potential and full life-cycle management requirements to drive innovation and achieve efficiency improvement, cost savings and risk reduction (Bertoni and Bertoni, 2022; Karagiannis et al., 2022). In a DT system, data collection, archiving and parsing are critical. Data serves as the core element, sourced from external databases, real-time system feedback, manual user inputs, deep integrations with third-party services, public information resources, as well as simulated and predicted data generated by advanced algorithms. In particular, the structure and type of the input and output should be consistent.

DT models achieve high-precision replication and simulation throughout the entire product lifecycle, ultimately providing accurate and timely decisions (Mohammadi et al., 2023). A typical narrative framework of DT modelling includes the generalised layered model, the asset administration framework, the decomposition and composition method, and the standardised architecture (Hu et al., 2023b). Verification and evaluation ensure that the DT model can truly reflect the structure, function and behavioural logic of a DT system (Seidel et al., 2023). In addition, potential mistakes and defects can be duly discovered and repaired to improve the reliability and stability of a DT system (Lugaresi et al., 2023). Data synchronisation and interaction ensure consistency between virtual and physical entities, for achieving seamless integration between them. Furthermore, data integration and interchange facilitate collaboration among different departments to enhance the overall productivity of the enterprise.

While data represents raw, unstructured inputs sourced directly from physical systems through sensors and IoT devices, information emerges as a higher-order construct derived from the contextualisation, processing and interpretation of that data, transforming it into actionable

insights. At the core of a DT system are the models, which are composed of complex mathematical, physical and analytical frameworks that serve as the system's computational foundation. The synthesis of data and information through these models creates a closed-loop feedback mechanism, enabling continuous refinement and synchronisation between the physical and virtual counterparts. This dynamic interaction allows for a deeper understanding of complex systems, the anticipation of future trends, the optimisation of design plans, virtual testing, and more (Ma et al., 2024).

DTs play several essential roles in advancing modern industries. They provide real-time monitoring, offering continuous insights into system performance and early detection of anomalies. These capabilities make DT technology vital for improving performance, reliability and innovation across various sectors.

1.3. Milestones in digital twin research

Since the introduction of the initial concept of a DT in 2002 by Michael Grieves in the context of product lifecycle management (PLM), the DT paradigm has undergone significant innovations and passed several milestones (Grieves and Vickers, 2016). These milestones are also graphically illustrated in Fig. 3.

Even though the concept was introduced in 2002, the term “digital twin” was only coined in 2010 by John Vickers (Grieves, 2014), prior to which it was referred to as the *Information Mirroring Model*. In 2011, in the first well-known application of this concept, the Air Force Research Laboratory (AFRL) utilised a DT for fighter aircraft maintenance. Subsequently, DTs were recognised as a key technology by both the United States (US) Air Force and the National Aeronautics and Space Administration (NASA). 2014 saw the publication of the first DT white paper titled “Digital Twin: Manufacturing Excellence through Virtual Factory Replication” (Grieves, 2014). Meanwhile, the DT paradigm was starting to be embraced and promoted by several industry leaders, including General Electric (GE), Dassault, and Siemens. For example, in 2015, GE utilised DTs for real-time monitoring and predictive maintenance of aviation engines. Similarly, Siemens launched the MindSphere system in 2016, which is based on DTs in Industry 4.0. Between 2017 and 2019, the American technological consulting firm Gartner consistently ranked DTs as one of the top ten emerging technologies (Jacoby and Usländer, 2020). The ISO/IEC 30173: 2023 international standard was released in 2023 (Gregory et al., 2024). Nowadays, DTs and related technologies are more widely adopted by industries to facilitate digital transformation (Tuegel et al., 2011), as well as to enhance their efficiency, flexibility and profitability (Böttjer et al., 2023).

During this time, several pioneering papers have been published regarding DTs in the industrial as well as new application domains (see Fig. 4). The first seminal paper was published by Tuegel et al. in 2011 about the application of DT for structural life prediction of aircraft (Tuegel et al., 2011). Between 2013 and 2017, several seminal papers were published, mainly focusing on the manufacturing (Reifsnider and Majumdar, 2013), automotive (Dattatreya et al., 2014) and energy (AS

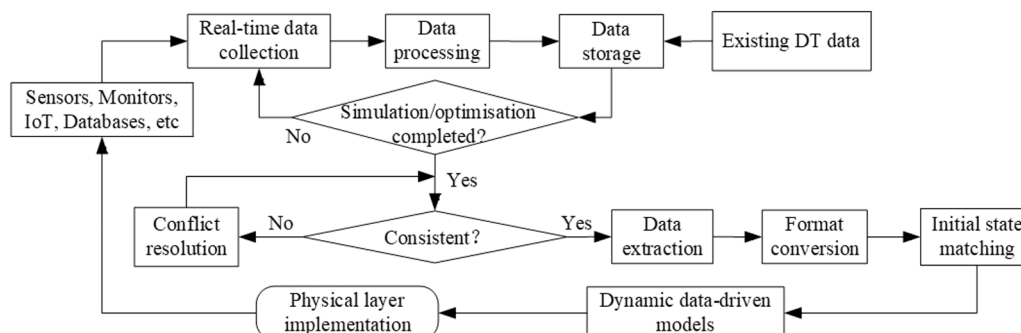


Fig. 2. A typical DT system's workflow.

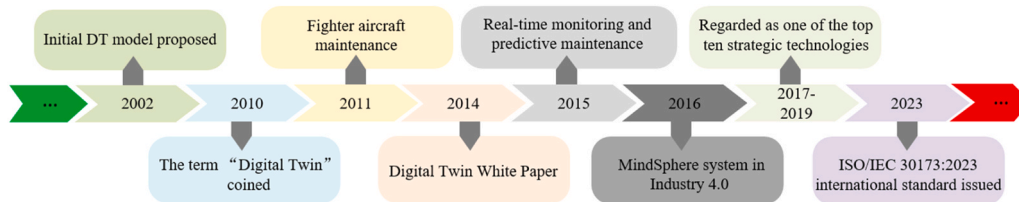


Fig. 3. Milestones of DT development.



Fig. 4. Timeline of the initiation of DT research across different industries.

and Antony, 2017) industries. Since 2018, there has been a significant growth in the interest of DT across more sectors, which is reflected in the sharp increase of related conference papers and journal articles (Ruohomäki et al., 2018; Jo et al., 2018; Korth et al., 2018; Naplekov et al., 2018; Rocca et al., 2020; Saini et al., 2018; Winning and Solutions, 2018). Currently, DT technology has established itself as one of most popular research topics for real-time simulation and optimisation in various industries (Semeraro et al., 2021).

The objective of this study is to conduct an in-depth analysis of articles published in the CII journal and categorise them by research domains in order to assess the research gaps in various fields. By mapping the landscape of existing studies, this review identifies areas of saturation as well as under-explored topics, ultimately highlighting key opportunities and challenges for future research. This analysis will guide scholars towards impactful research directions and promote further innovation in the identified fields.

The rest of the paper is organised as follows. Section 2 defines the review protocol. Section 3 provides a detailed literature review of DT applications published in the CII journal. Section 4 discusses research opportunities of DT. Finally, Section 5 highlights future challenges and managerial insights.

2. Review protocol

A review protocol was designed to systematically guide the entire review process and a rigorous quality assessment was conducted to ensure the inclusion of the most recent advancements and critical issues in DT research. First, we outline the search strategy used to identify and select DT-related articles, detailing the inclusion and exclusion criteria. Next, we define the scope of the review, focusing specifically on DT-related studies published in CII. We also analyse the trends in DT-related publications, including all the selected articles in addition to only those published in the CII journal.

2.1. Search strategy

In this study, we followed the literature review methodology of Ashraf et al. (2016), Brereton et al. (2007), Keele (2007), Saeed et al. (2022) and Singh et al. (2021) to ensure a comprehensive and unbiased review of the relevant DT literature. The search terms and strings were carefully designed to align with the research questions (Dieste et al., 2009). We explored relevant publications in the Web of Science (WoS) database. The focus was on identifying recent articles with the keyword “digital twin” in the “Topic” field. Additionally, to ensure the relevance to industrial applications and to explore how the DT technology is applied across various industrial sectors, the search also included

articles that contain keywords starting with “industr*” (e.g., industry, industrial and industrialised) in the “Topic” field.

2.2. Inclusion/exclusion criteria

We followed the inclusion/exclusion criteria, as well as the quality assessment process, outlined by Ebrahimighahnavieh et al. (2020) in selecting the reviewed papers. The review was conducted up to the end of 2024, resulting in an initial pool of 5084 articles.

For the inclusion of articles in this review, we applied the following criteria:

- **Journal Impact Factor:** Only articles published in journals with an impact factor of 5 or higher were included. This threshold ensures that the selected studies come from high-quality and reputable sources.
- **Language:** Only articles published in English were included to maintain consistency in the analysis and ensure content accessibility.
- **Article Type:** Both research articles and review articles were included, as they provided original research findings and comprehensive overviews of existing knowledge.

After applying these criteria, the initial pool of 5084 articles was reduced through a preliminary screening process to 1441 articles that met the basic quality standards. A further in-depth evaluation resulted in an even smaller pool of final articles that were deemed most relevant for the subsequent literature review. This approach resulted in a focused, high-quality set of literature to address our research objectives.

Table 1 presents the top 30 journals ranked by the number of papers published from our search results. The table provides details of the journals, their publishers, the number of publications and the publication years. According to the number of publications, the *Journal of Manufacturing Systems* is the leading contributor among the top 30 journals, followed by CII. Notably, the *Journal of Manufacturing Systems*, CII, *IEEE Transactions on Industrial Informatics* and *Journal of Cleaner Production* published their first articles on DT as early as 2018, demonstrating their high level of awareness and responsiveness to emerging trends in this field.

The reviewed journals fall into the fields of operations research, mathematics, computer science, engineering, building, medical and healthcare. The *Journal of Manufacturing Systems*, CII and *Robotics and Computer-Integrated Manufacturing* share a common focus on advancing DT technology in the manufacturing sector. They emphasise intelligent, data-driven optimisation and the integration of cyber-physical systems, along with promoting effective human-machine collaboration. Collectively, these journals explore how DT technology can boost

Table 1
Leading journals contributing to DT research.

Journals	Publishers	Numbers	Publication years
Journal of Manufacturing Systems	Elsevier	138	2018–2024
Computers in Industry	Elsevier	80	2018–2024
Robotics and Computer-Integrated Manufacturing	Elsevier	79	2019–2025
Advanced Engineering Informatics	Elsevier	74	2019–2024
Automation in Construction	Elsevier	72	2019–2024
International Journal of Production Research	Taylor & Francis	63	2019–2024
Journal of Intelligent Manufacturing	Springer Nature	57	2019–2024
IEEE Transactions on Industrial Informatics	IEEE Institute of Electrical and Electronics Engineers	53	2018–2024
Computers & Industrial Engineering	Elsevier	42	2020–2024
IEEE Internet of Things Journal	IEEE Institute of Electrical and Electronics Engineers	33	2020–2024
Journal of Cleaner Production	Elsevier	30	2018–2024
Journal of Industrial Information Integration	Elsevier	29	2020–2024
Measurement	Elsevier	23	2021–2024
Mechanical Systems and Signal Processing	Elsevier	20	2020–2025
Journal of Building Engineering	Elsevier	19	2020–2024
Journal of Manufacturing Processes	Elsevier	18	2020–2024
Engineering Applications of Artificial Intelligence	Elsevier	17	2021–2024
Future Generation Computer Systems	Elsevier	17	2020–2025
IEEE Transactions on Automation Science and Engineering	IEEE Institute of Electrical and Electronics Engineers	16	2022–2024
Energy and Buildings	Elsevier	15	2019–2024
Engineering Structures	Elsevier	13	2021–2024
International Journal of Production Economics	Elsevier	12	2021–2024
Expert Systems with Applications	Elsevier	12	2021–2024
Applied Energy	Elsevier	11	2022–2024
Energy	Elsevier	11	2021–2024
IEEE Communications Surveys and Tutorials	IEEE Institute of Electrical and Electronics Engineers	10	2021–2024
Reliability Engineering & System Safety	Elsevier	10	2022–2024
IEEE Transactions on Instrumentation and Measurement	IEEE Institute of Electrical and Electronics Engineers	10	2022–2024
Sustainable Cities and Society	Elsevier	10	2022–2024
Internet of Things	Elsevier	9	2023–2024

Note: While the search was conducted up to the end of 2024, some papers already have the publication year as 2025.

manufacturing efficiency, enable seamless interactions between virtual models and physical systems, and optimise production processes.

However, notable differences exist in their research foci. The *Journal of Manufacturing Systems* is primarily concerned with applications within manufacturing enterprises, such as process planning, real-time monitoring and operational optimisation. In contrast, CII adopts a broader perspective, exploring DT applications across the industrial ecosystem, including supply chain management and inter-enterprise integration. Meanwhile, *Robotics and Computer-Integrated Manufacturing* focuses on

the role of DT in robotics and automated manufacturing systems, with particular emphasis on the optimisation of intelligent robotic units and production lines. These differences in the research scope, application focus and technical depth contribute to a complementary landscape, enriching the field of DT research through diverse perspectives and methodologies.

2.3. Popularity analysis

The popularity of DT research and publications is analysed in Fig. 5, focusing on the publication timeline and annual volume of selected articles. Since 2017, when the first DT-related article was published in the *Scripta Materialia* journal, the number of published articles has steadily increased. Starting in 2020, the total number of articles has sharply increased. Notably, by the end of 2024, articles scheduled for publication in 2025 have already emerged, indicating the sustained and growing interest in DT research.

The analysis also examines publication trends within CII. The first DT-related article by CII was published in May 2018, titled “Automatic assembly planning based on digital product descriptions”. In 2019, the journal released a special issue, titled “Digital transformation of manufacturing through cloud services and resource virtualisation”, which explicitly included “digital twin” as a keyword. This special issue highlighted how cloud services facilitate the digital transformation of manufacturing, enhancing the integration between digital and physical systems in the context of Industry 4.0 (Borangui et al., 2019). Since 2018, DT-related publications in CII have consistently increased, peaking in 2023 before a slight decline in 2024. Further analysis of CII publications reveals the impact of including or excluding the term “industr*” in the search criteria. When both “digital twin” and “industr*” were used as keywords, 68 articles were identified, focusing on the application of DT in industrial contexts. Removing the “industr*” constraint, while retaining “digital twin”, resulted in an increased count of 81 articles. One of the articles is an editorial authored by the guest editors for a special issue. The additional articles mainly address research on technical innovations and management paradigms, suggesting that while DT technology finds increasing applications in industrial domains, research is also advancing in broader aspects, such as improving the underlying technologies and exploring new management approaches. The overall trend indicates that, as DT technology continues to be adopted across various sectors, there is a parallel emphasis on advancing the technology to ensure its continuous evolution and adaptability in response to emerging challenges and opportunities. Focusing on both applications and innovation underscores the growing recognition of DT as a crucial element of modern manufacturing and industrial strategies.

The data is up to date until the end of 2024, and it shows an obvious decline in the number of DT-related articles published in CII in that year. As shown in Fig. 6, our analysis of CII publications from 2017 to 2024 reveals that 2024 has the lowest publication count for the past several years. The decrease in DT-related publications in CII in 2024 could well just be a temporary fluctuation or may be due to changes in editorial priorities or submission trends, though the underlying causes remain to be further examined. Despite this decline, DT-related articles still account for a substantial proportion, ranking as the second-largest category among all CII publications during this period. Although the overall number of DT-related papers has increased, CII continues to show strong interest in DT research.

2.4. Quality assessment criteria

It is essential to assess the quality of a review, which helps evaluate different studies and offer insightful suggestions for further research (De Beer et al., 2014). Our quality assessment criteria are related to the article types, the published year and the impact factor. The articles selected for this review closely align with the research domains covered

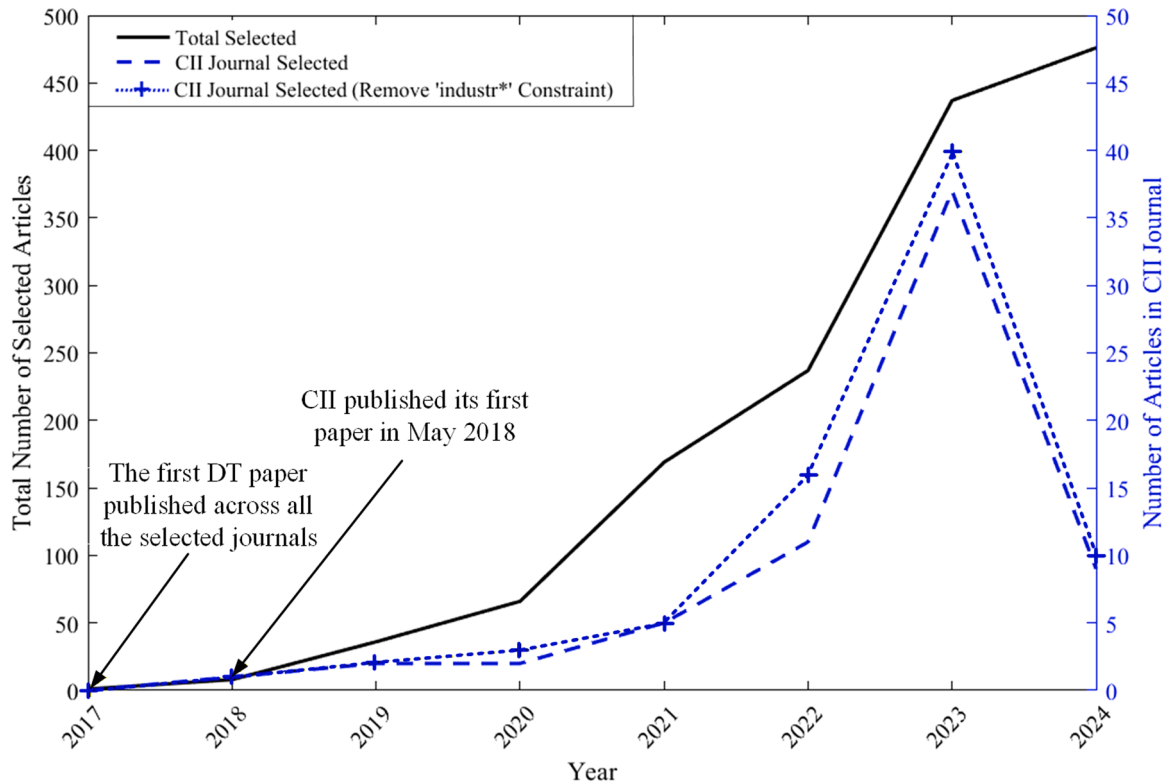


Fig. 5. Publication trend in selected journals (2017–2024).

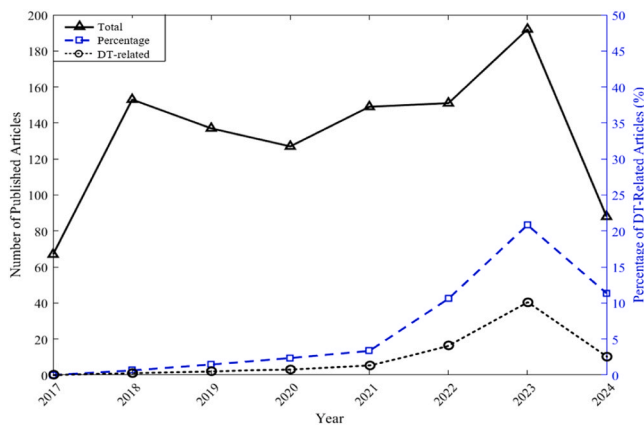


Fig. 6. Publication trend in the CII journal (2017–2024).

in the CII journal. Furthermore, all of these articles are published in high-impact journals (impact factor > 5), and their publication dates (post-2017) ensure both the rigour and contemporary relevance required for a comprehensive analysis. A word cloud visualisation was generated from the titles of the selected articles to visualise the key themes and industries reviewed. As shown in Fig. 7(a) and (b), word clouds were generated from the titles of all selected articles and only the CII journal, respectively. Fig. 7(b) shows two distinct zones. The inner area (inside the ellipse), formed by 68 article titles, represents topics containing both “digital twin*” and “industr*”. The outer area (outside the ellipse), consisting of 81 article titles, represents topics mentioning only “digital twin*”.

The broader literature from the WoS database highlights a diverse range of application beyond manufacturing, including sustainability, energy management and blockchain integration. This showcases the interdisciplinary potential of DT. The articles published in CII primarily

concentrate on industrial applications, with an emphasis on integrating DT into manufacturing systems, reflecting a strong focus on practical implementation within Industry 4.0. This analysis illustrates the evolution of DT research, from its industrial roots towards a more interdisciplinary and diverse application landscape.

3. Literature review

Our review begins by systematically analysing all DT-related articles published in the CII journal, while also aiming for a comprehensive understanding of the broader DT research published by other journals. The initial set of 80 (see Section 2.2) articles published between 2018 and 2024 was further supplemented by 4 recent publications from early 2025, resulting in a total of 84 articles that were selected for in-depth analysis. Through a comprehensive analysis of these 84 DT-related articles, we have meticulously classified the research into four primary domains: manufacturing, construction, and transportation industries; as well as technologies and paradigms, as illustrated in Fig. 8. For each domain, we offer a detailed analysis of the current state of knowledge and its developmental trajectory. Following this, we conducted a critical evaluation of the key contributions within each category, thoroughly examining the methodologies, findings and theoretical implications.

3.1. Manufacturing

The following is a review of research on DT applications in manufacturing. CII, as a leading journal that bridges academic research and practical industrial applications, has published 48 DT-related papers in the manufacturing sector. We have categorised these articles into five main groups, which are zero defect manufacturing and quality management, process optimisation and automation, predictive maintenance and health management, architectures and adaptive systems, and integration with advanced technologies, as shown in Table 2.

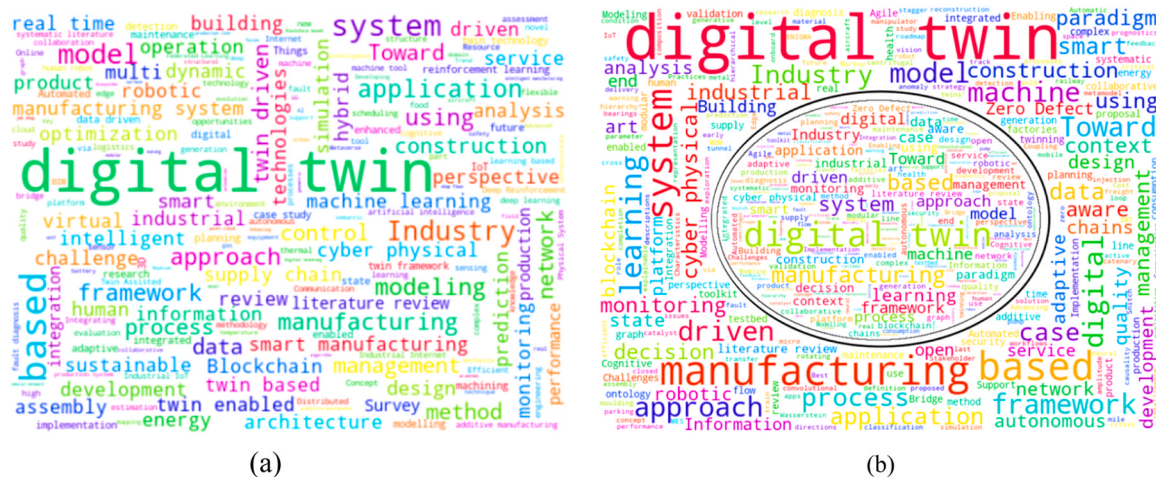


Fig. 7. (a) All selected; (b) CII journal selected. Word clouds generated by article titles.

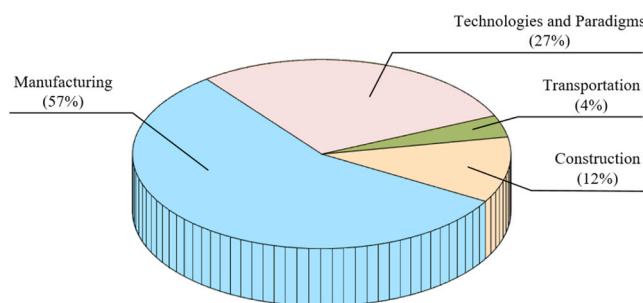


Fig. 8. Distribution of CII journal publications across different industries (2018–2025).

3.1.1. Zero-defect manufacturing and quality management

DTs provide innovative solutions for quality management, zero-defect manufacturing and zero-waste initiatives. [Pсарommatis et al. \(2023\)](#) proposed an ontology-driven approach to integrate DT technology into zero defect manufacturing ecosystems, offering a formalised method for aligning multiple systems and enhancing quality management in Industry 4.0 settings. [Christou et al. \(2022\)](#) introduced an IoT-based platform for Quality 4.0, integrating predictive maintenance and zero-defect manufacturing with advanced machine learning and big data analytics to enhance production efficiency and reduce product defects in industrial environments. [Calvo-Bascones et al. \(2023\)](#) proposed a novel anomaly detection methodology within a DT ecosystem, termed the Snitch Digital Twin (SDT). This approach modelled the interrelationships among behaviours in industrial systems to enhance anomaly detection capabilities, specifically applied to a diesel generator cooling system. The SDT framework extends beyond conventional DT applications by focusing on the characterisation of interactions between physical behaviours, leveraging spatiotemporal features such as density distribution, trend slopes and intercept values obtained via sliding time windows. The proposed methodology is effective in reducing false positives and providing a more nuanced characterisation of anomalous behaviours in dynamic environments.

DTs have also been applied to supply chain management to foster broader goals of waste reduction and quality enhancement. [Sengupta and Dreyer \(2023\)](#) explored the use of DTs to drive zero-waste value chains in grocery retail, extending the zero-defect manufacturing mindset to supply chain optimisation and waste reduction. Data and models work together to build a virtual replica that reflects the state and behaviour of the physical entity in real time. [Isaja et al. \(2023\)](#) introduced a blockchain-based framework to ensure trusted quality data

sharing in zero defect manufacturing, enabling secure horizontal and vertical data integration across supply chains to drive zero waste value chains. Ensuring the accuracy of DT models is foundational to realising their intended benefits. [Lugaresi et al. \(2023\)](#) introduced a novel approach for online validation of DT in manufacturing, utilising similarity measures between physical and virtual systems to ensure model accuracy and support short-term decision-making in dynamic production environments. The implementation of zero-defect manufacturing through DT relies on technological integration and theoretical innovation. [Powell et al. \(2022\)](#) presented an overview of contemporary advancements in zero defect manufacturing, highlighting emerging trends in defect prevention. They also proposed future research directions, particularly towards achieving zero-waste value chains.

3.1.2. Process optimisation and automation

DT technology has been increasingly applied across various domains to enhance automation, optimisation and decision-making processes in both manufacturing and business environments. [Garcia et al. \(2025\)](#) developed a physics-informed DT framework for optimising process settings in continuous manufacturing, emphasising interpretability and human-centric design, though limited by its focus on steady-state behaviour. [Perno et al. \(2023\)](#) introduced a machine learning-enabled DT framework to predict and optimise critical process parameters in catalyst manufacturing, improving process reliability and operational efficiency. [Modoni et al. \(2022\)](#) presented a DT framework for real-time in-process monitoring and optimisation in micro-injection moulding, improving process reliability and product quality in micro-manufacturing. [Hu et al. \(2023b\)](#) proposed a Petri nets-based DT framework to improve coordination in dual-arm robotic systems, enhancing their capability for complex, cooperative industrial tasks. [Sierla et al. \(2018\)](#) presented a DT-driven approach for product-centric assembly planning, enabling flexible manufacturing systems that minimise reconfiguration needs and optimise resource orchestration. [Lugaresi and Matta. \(2023\)](#) introduced a method for automated DT generation using object-centric process mining, providing a robust solution for modelling complex material flows in assembly lines and enhancing decision-making in manufacturing environments.

Current research on DT technology primarily focuses on optimising individual stages or limited areas of the production process. However, the generalisation capability of these models across different manufacturing processes or industries still requires further validation. Additionally, the real-time responsiveness of DT and its integration with other systems require further improvement. This is particularly important for optimising end-to-end production scheduling, cross-departmental resource allocation and cross-industry applications.

Table 2

Summary of DT applications in the manufacturing industry as published in the CII journal.

Categorisation	Specific Focus	Refs.
Zero Defect Manufacturing and Quality Management	Ontology-driven zero-defect manufacturing integration	(Psarommatas et al., 2023)
	Quality management and zero-defect manufacturing	(Christou et al., 2022)
	Behavioural anomaly detection	(Calvo-Bascones et al., 2023)
	Food waste reduction	(Sengupta and Dreyer, 2023)
Process Optimisation and Automation	Blockchain for secure and reliable data sharing	(Isaja et al., 2023)
	Online validation of simulation-based DT	(Lugaresi et al., 2023)
	Defect prevention and compensation	(Powell et al., 2022)
	Physics-informed DT for process setting selection	(Garcia et al., 2025)
	Prediction and optimisation of critical process parameters	(Perno et al., 2023)
	Optimisation of micro-injection moulding	(Modoni et al., 2022)
	Petri nets for dual-arm robotic coordination	(Hu et al., 2023b)
	Product-centric assembly planning	(Sierla et al., 2018)
	Automating the management of complex material flows	(Lugaresi and Matta, 2023)
	MES/MOM implementation for IT/OT integration	(Mantravadi et al., 2023)
	Automatic conversion of fluid circuit diagrams	(Stegmaier et al., 2024)
	Surface mount technology	(Seidel et al., 2023)
	Data-driven production monitoring	(Xu et al., 2024a)
	Business process management optimisation	(Beerepoot et al., 2023)
Predictive Maintenance and Health Management	Unsupervised fault detection of rolling bearings	(Zhang et al., 2025)
	Scalable real-time monitoring platforms	(Tapia et al., 2024)
	Prognostics and health management	(Toothman et al., 2023)
	Part qualification and defect management	(Phua et al., 2022)
	Centrifugal pump fault diagnosis	(Xu et al., 2024b)
	Condition-based maintenance	(Marquez et al., 2023)
	Knowledge graph-driven model updates	(Su et al., 2024)
	Optimising product quality control	(Caiazzo et al., 2022)
Integration with Advanced Technologies	Optimising energy usage and monitoring machine health	(Pantazis et al., 2023)
	Energy consumption monitoring and sustainability	(Abdoun et al., 2023)
	Simulation of performance and behaviour	(Pires et al., 2023)
	Optimising task assignment	(De Giacomo et al., 2023)
	Ontology-based data representation	(Rico et al., 2023)
	Product and service life cycle management	(Woitsch et al., 2022)
	Cybersecurity threat modelling	(Jbair et al., 2022)
	Blockchain for secure and traceable production processes	(Mandolla et al., 2019)
	Adaptive scheduling	(Müller-Zhang et al., 2023)
	Real-time adaptive DT architecture	(Ogunsakin et al., 2023)

Table 2 (continued)

Categorisation	Specific Focus	Refs.
	Modular DT for agile machine development	(de Oliveira Hansen et al., 2024)
	AR-based interactive visualisation and deformation tracking	(Liao et al., 2025)
	Real-time optimisation and decision-making	(Albini et al., 2023)
	Bridge asset monitoring and management	(Mohammadi et al., 2023)
	Adaptive DT patterns for Industry 4.0	(Bellavista et al., 2023)
	Real-time monitoring and reconfigurable manufacturing	(Friederich et al., 2022)
	Lifecycle management	(Bertoni and Bertoni, 2022)
	Literature review on DT paradigms and challenges	(Semeraro et al., 2021)
	Roadmap for autonomous DT	(Hribernik et al., 2021)
	Well-defined services	(Cimino et al., 2019)
	Active learning techniques	(Chabanet et al., 2021)
	Adaptive holonic manufacturing	(Valckenaers, 2020)

Mantravadi et al. (2023) proposed a framework for integrating Manufacturing Execution Systems (MES)/Manufacturing Operations Management (MOM) with Industrial IoT in smart factories, highlighting design recommendations to enhance manufacturing flexibility and reconfigurability across Industry 4.0 supply chains. The potential impact of integrating multiple subsystems and devices on system response speed has also not been thoroughly investigated, particularly regarding real-time performance in complex industrial environments. Stegmaier et al. (2024) proposed a method for the automated digitalisation of fluid circuit diagrams using object detection and image processing techniques. This approach extracts symbols and connections from the diagrams and converts them into digital formats to build a DT of fluid systems. It is particularly effective for high-quality images, significantly reducing manual efforts and speeding up the digitalisation process. The capability to handle more complex low-quality images or non-standardised charts has not been thoroughly explored.

The effectiveness and robustness of automated processing still require further improvement. Seidel et al. (2023) introduced a DT framework that integrates machine learning models for optimising the surface mount technology manufacturing process, improving product quality and reducing production costs through structured data acquisition. While structured data collection holds potential for optimising manufacturing processes, issues such as data breadth, quality, real-time availability and security have not been thoroughly addressed in discussions on the effectiveness of DT applications. Xu et al. (2024a) presented a data-driven DT approach for comprehensive performance monitoring in production lines. This approach utilises probabilistic models and logistic regression to predict deviations from production targets, enhancing both machine and system-level efficiency. In environments with scarce or low-quality data, the effectiveness and robustness of the system may be compromised. DT applications are not limited to physical manufacturing processes but also extend to business process management. Beerepoot et al. (2023) explored the role of DT in business process management, emphasising their potential to automate, optimise and predict business processes while addressing integration challenges with existing models.

3.1.3. Predictive maintenance and health management

The application of DT technology in predictive maintenance and health management spans across various sectors, including aerospace manufacturing, mechanical engineering and additive manufacturing.

Zhang et al. (2025) proposed a gradient fusion DT for rolling bearing fault detection using only healthy data, effectively addressing label scarcity; however, its performance in noisy or real-time environments remains to be validated. Tapia et al. (2024) developed a scalable software platform for real-time monitoring of machine tools, employing big data techniques and distributed computing to detect outliers and visualise process data. This platform, designed for aeronautical manufacturing, demonstrated effective performance in a real machining environment and scalability through simulated tests. Toothman et al. (2023) proposed a DT framework for standardising prognostics and health management in manufacturing, allowing reusable models for fault detection and predictive maintenance. This framework was demonstrated through a case study on industrial pump systems. Although the standardised DT framework offers a certain level of generalisability, its adaptability and scalability remain limited in different industries and application scenarios. Phua et al. (2022) introduced a hierarchical DT framework for metal additive manufacturing, utilising machine learning and in-situ sensing for real-time quality control, part qualification and process optimisation. The complex physical and chemical reactions in the additive manufacturing process are difficult to fully capture in real time using existing DT models. The application of DT technology often relies on large volumes of high-quality data, but in real industrial settings, data scarcity or incompleteness remains a major bottleneck. Xu et al. (2024b) presented a novel DT approach for centrifugal pump fault diagnosis, leveraging transfer learning and graph convolutional networks to address challenges in fault detection with minimal data, significantly improving operational reliability. Achieving data interoperability across different manufacturing stages is essential for forming a truly closed-loop intelligent manufacturing system. Marquez et al. (2023) demonstrated the use of DT in condition-based maintenance of train axle bearings, showcasing predictive models and real-time monitoring to enhance asset longevity and reduce operational downtime. Su et al. (2024) proposed a knowledge-driven DT system for manufacturing, utilising knowledge graphs for continuous process optimisation and decision-making, verified through its application in an aero-engine blade production line. Caiazzo et al. (2022) proposed zero defect manufacturing strategies utilising DTs to improve detection, prediction and prevention techniques in manufacturing, highlighting key technological challenges and future research directions in the quest for zero defects in Industry 4.0 environments. Pantazis et al. (2023) introduced a virtual machining toolkit that predicts computer numeric control milling machine power consumption, leveraging DT technology to optimise energy use and monitor machine health. However, extending the use of DT to global energy consumption management and health monitoring can maximise the potential benefits.

3.1.4. Integration with advanced technologies

The DT technology enhances real-time monitoring, task optimisation, decision support and cybersecurity by integrating with emerging technologies such as data-driven models, reinforcement learning, Markov decision processes and blockchain. Abdoune et al. (2023) proposed a data-driven approach to integrate energy consumption models into DTs, enabling real-time energy monitoring and contributing to sustainable manufacturing practices. Data-driven models often rely on accurate real-time data collection. If the data quality is poor or system integration is incomplete, the monitoring effectiveness can be significantly impacted. Pires et al. (2023) introduced a trustworthy reinforcement learning model integrated with DT technology for manufacturing decision support, enhancing scenario prediction accuracy and reducing decision-making time under data sparsity and cold-start issues. The application of reinforcement learning in complex industrial scenarios faces challenges of high computational cost and long training time, particularly with regard to improving adaptability in dynamic environments. Effectively processing diverse types of manufacturing data (e.g., images, sensor data, etc.) and ensuring model interpretability remain critical challenges for realising industrial applications. De Giacomo et al.

(2023) applied Markov decision processes to DT composition in smart manufacturing, optimising task assignment and manufacturing workflows under uncertainty to improve cost and quality outcomes. Markov models are highly dependent on the state space, and in complex, high-dimensional manufacturing environments, the issue of state explosion may hinder their efficiency and scalability in practical applications.

Rico et al. (2023) highlighted the use of ontology networks to achieve context-aware data representation in DTs, enabling better interoperability and decision-making in digital factories by ensuring syntactic and semantic alignment of shared data. The construction and maintenance of ontology networks are highly complex, especially when dealing with large volumes of heterogeneous data sources. Ensuring dynamic updates and scalability of the ontology presents a significant challenge. Woitsch et al. (2022) introduced a model-based approach using DT technology to bridge data integration gaps in the product-service life cycle, demonstrating its effectiveness in improving sterilisation and data-driven decision-making in manufacturing systems. Handling various data types generated at different stages of the lifecycle while ensuring data quality remains an issue that requires further investigation.

Jbair et al. (2022) presented a DT-integrated structured threat modelling approach to predict and mitigate cybersecurity risks in industrial cyber-physical systems, enhancing the resilience of smart manufacturing systems. The diversity and dynamic nature of complex network attacks necessitate real-time updates to threat models. Additionally, cybersecurity measures may increase system complexity, making it crucial to study how to balance security and system efficiency. Mandolla et al. (2019) presented a DT model for additive manufacturing in the aircraft industry, leveraging blockchain technology to ensure transparency, security, and traceability in production, providing a solution for managing complex supply chains. The introduction of blockchain may increase system complexity and computational overhead, especially in large-scale production and real-time processing, where scalability and computational costs could become bottlenecks. Additionally, effectively handling non-standardised data in additive manufacturing processes presents a significant challenge for this approach.

3.1.5. Architectures and adaptive systems

DTs' architectures and adaptive systems in manufacturing focus on enhancing scalability, flexibility and real-time adaptability for optimising production processes within dynamic industrial environments. Müller-Zhang et al. (2023) explored the integration of DT with deep reinforcement learning, enabling dynamic and adaptive scheduling for service-based production systems. Deep reinforcement learning incurs high computational costs, particularly when handling large-scale industrial data and diverse production processes, making the efficiency of model training and updates a challenge. Additionally, the low interpretability of reinforcement learning models may affect the transparency and reliability of decision-making in industrial applications. Ogunsakin et al. (2023) introduced a real-time adaptive architecture for DT technology, facilitating dynamic reconfigurations and online optimisations in response to continuous changes in production environments. The research focuses on flexibility and reconfiguration in real-time production environments, making it suitable for dynamic and constantly changing production settings. However, there is limited evaluation of long-term system performance, especially in highly complex and frequently changing production processes. The scalability and stability of such systems require further investigation.

de Oliveira Hansen et al. (2024) proposed a framework for agile digital machine development using modular DT designs, which enable reconfigurability and rapid adaptation in mechatronic discrete manufacturing systems, significantly reducing time to market and enhancing system flexibility. Modular design enables rapid adaptation of manufacturing systems, but its effectiveness in terms of scalability

across industries or domains has not been fully explored, particularly when production systems become more complex or heterogeneous. [Liao et al. \(2025\)](#) introduced an augmented reality-based DT for visualising soft robot deformations; this offers intuitive interaction but lacks an in-depth exploration of autonomous control or deployment robustness. [Albini et al. \(2023\)](#) presented a real-time simulation and optimisation algorithm for a DT of a robotic cell, integrating human operator behaviours to improve context-awareness and adaptability in manufacturing environments. The framework’s ability to adapt to different types of operator behaviours, as well as the complexity of acquiring and analysing behavioural data, has not been sufficiently discussed. Additionally, maintaining a balance in human-machine collaboration in highly automated scenarios remains a key challenge. [Mohammadi et al. \(2023\)](#) presented a bridge information modelling-based methodology that integrates terrestrial laser scanning data with decision support systems to enhance bridge management and maintenance, validated through a real-world application on the Werrington Bridge. The scalability of this approach in other infrastructure management contexts warrants further exploration. [Bellavista et al. \(2023\)](#) defined design patterns for developing adaptive and context-aware DTs in Industry 4.0, focusing on microservices architecture to meet the demands of dynamic production environments. The management costs of complex system integration and service communication may become a barrier in large-scale industrial applications, especially in highly distributed systems. Coordinating and scheduling services could increase system vulnerability and latency.

[Friederich et al. \(2022\)](#) proposed a DT-based data-driven framework in smart manufacturing, focusing on automating the generation and validation of simulation models. It utilises machine learning and process mining to support reconfigurable manufacturing and real-time monitoring, enhancing model adaptability and accuracy. This approach excels in model generation and real-time monitoring but relies heavily on large amounts of high-quality historical data. If data quality is poor or data is scarce, the robustness and adaptability of the model may be limited. Therefore, improvements are needed in handling data scarcity and noisy data. [Bertoni and Bertoni \(2022\)](#) explored the role of DT in Product-Service System (PSS) design, highlighting their ability to predict hardware availability and support solution development. It identifies challenges in achieving real-time integration between physical and virtual systems, offers a clearer definition of PSS DT and recommends research to improve data flow, synchronisation, and lifecycle management. The real-time integration of physical and virtual systems, particularly in data flow management, synchronisation and lifecycle management, remains a major challenge for comprehensive lifecycle management. [Semeraro et al. \(2021\)](#) explored the DT paradigm, identifying its main components and challenges across industries, offering a comprehensive analysis of ongoing research and technical difficulties, particularly in predictive manufacturing and cyber-physical systems. Although the research demonstrates the potential of DT across industries, it lacks specific solutions. Addressing real-time performance and integration challenges in industrial applications requires further investigation. [Hribernik et al. \(2021\)](#) explored the evolution of DT towards autonomy, context-awareness and adaptivity, identifying research gaps and proposing a roadmap for the realisation of fully autonomous DT in future smart factories. The study outlined a comprehensive roadmap for the future development of DT technology. [Cimino et al. \(2019\)](#) synthesised DT applications in manufacturing, identifying gaps between theoretical capabilities and practical implementations, particularly in relation to manufacturing execution systems and real-time integration. The research focused on current implementation challenges, aiming to bridge the gap between theory and practice. [Chabanet et al. \(2021\)](#) proposed an active learning-based method that integrates digital simulations and machine learning meta-models for smart factory applications, enabling efficient model updates and improving decision-making processes. [Valckenaers \(2020\)](#) explored the evolution of holonic manufacturing systems, transitioning from the

PROSA framework to ARTI. The study highlighted the role of DT in creating adaptive, intelligent manufacturing environments that integrate dynamic decision-making and flexibility. In practical applications, attention must be given to data quality and integration issues.

3.1.6. Critical analysis and summary

The application of DTs in manufacturing spans quality management, process optimisation, predictive maintenance, and integration with advanced technologies. Many studies demonstrated notable benefits, such as improved anomaly detection, real-time process control, and enhanced system flexibility. However, critical limitations persist. These include over-reliance on high-quality and structured data, limited generalisability across production settings, and insufficient integration of DT models with broader enterprise systems. Moreover, while advanced technologies like AI, blockchain, and semantic modelling strengthen autonomy and decision support, they also introduce challenges such as computational overhead, data interoperability issues, and reduced model interpretability. Collectively, these limitations highlight the need for more robust, scalable, and adaptive DT solutions in real-world manufacturing environments.

3.2. Construction

The CII journal has published 10 articles focusing on DT applications in the construction sector, and those articles have been systematically categorised into three key areas, namely monitoring and decision-making, human-robot collaboration and modular integrated construction, as illustrated in [Table 3](#).

3.2.1. Monitoring and decision-making

DTs are being increasingly applied for monitoring and decision-making in the construction industry. [Fawad et al. \(2025\)](#) developed an immersive DT platform combining Augmented Reality (AR) and Scan-to-Building Information Modelling (BIM) for real-time bridge damage assessment and model updates, enhancing structural monitoring and decision-making. Although the platform has demonstrated efficiency in structural monitoring, it is highly dependent on data quality. Further exploration is needed on how to handle incomplete or low-quality scan data. Additionally, the scalability and adaptability of the platform have yet to be validated in more complex bridge and infrastructure environments. [Xia et al. \(2023b\)](#) improved subsea robot teleoperation by integrating Virtual Reality (VR) with a DT, providing enhanced sensory feedback through visual and haptic inputs, leading to better operator awareness and control in complex underwater

Table 3
Summary of DT applications in construction industry as published in CII journal.

Categorisation	Specific Focus	Refs.
Monitoring and Decision-making	Bridge monitoring and management	(Fawad et al., 2025)
	Enhancing sensory feedback in subsea teleoperation	(Xia et al., 2023b)
	Uncertainty management in offshore wind turbines	(Jorgensen et al., 2023)
	A real-time safety warning system	(Ye et al., 2023)
	Logistics optimisation on construction sites	(Greif et al., 2020)
Human-robot Collaboration	Systematic review and conceptual clarification of HDTs in Industry 5.0	(Gaffinet et al., 2025)
	Automation in construction and facility management	(Zhang et al., 2022)
	Adaptive construction workflows	(Wang et al., 2024a)
Modular Integrated	Real-time monitoring, optimisation and automation of construction processes	(Jiang et al., 2022)
	Addressing information fragmentation	(Jiang et al., 2021)

environments. In more complex environments, enhancing system response speed and multimodal sensing is a meaningful area for further research. [Jorgensen et al. \(2023\)](#) outlined the essential requirements for implementing the DT paradigm in Offshore Wind Turbine (OWT) structures, particularly for analysing fatigue under uncertain conditions. It emphasises the need for capturing and managing uncertainties using Gaussian Process surrogate models to ensure reliable lifetime assessments and decision-making for OWT support structures. The accuracy and computational cost of surrogate models remain a challenge in large-scale applications, particularly in more complex or multivariable wind turbine environments. [Ye et al. \(2023\)](#) presented a DT framework for real-time safety monitoring and early warning during tunnel construction, demonstrating its effectiveness in predicting and mitigating safety risks in complex underground environments. [Greif et al. \(2020\)](#) investigated the potential of DT to improve logistics in construction, focusing on silo fill-level monitoring to optimise supply chains and reduce inefficiencies. The research primarily focuses on a single logistics aspect (i.e., warehouse monitoring), lacking analysis of more complex supply chain environments. Specifically, it does not address handling multi-node, multi-variable supply chain data or real-time responses to supply chain fluctuations.

3.2.2. Human-robot collaboration

Human-robot collaboration has emerged as a critical area of research due to the influence of human involvement on construction systems, with the goal of enhancing automation and reducing human-related inefficiencies. [Gaffinet et al. \(2025\)](#) systematically reviewed and clarified the concept of Human Digital Twins (HDTs), proposing a taxonomy and introducing the notion of augmented HDTs under the Industry 5.0 paradigm. However, the study offers only some limited discussion on practical implementation, ethical considerations, and real-time industrial applicability. [Zhang et al. \(2022\)](#) proposed a BIM-robot integration framework to streamline building automation and robotic tasks, focusing on lifecycle information flow and enhanced DT interoperability in the construction industry. The study emphasised the execution of robotic automation tasks, with a particular focus on managing lifecycle information flow. [Wang et al. \(2024a\)](#) introduced a closed-loop DT framework integrating BIM into human-robot collaborative workflows for construction, enhancing flexibility and accuracy in real-time task adaptation and execution. This study focused on human-machine collaboration workflows, aiming to enhance task execution flexibility and accuracy.

3.2.3. Modular integrated construction

Modular integrated construction has the potential to enhance efficiency, resource management and stakeholder collaboration. [Jiang et al. \(2022\)](#) presented a DT-enabled smart modular integrated construction system, improving on-site assembly through real-time monitoring and enhanced resource management for higher efficiency and safety. The study lacked an in-depth discussion on real-time data exchange between multiple stakeholders, particularly in cross-departmental collaboration and complex supply chain environments. [Jiang et al. \(2021\)](#) proposed a blockchain-enabled platform for integrating DT technology in modular integrated construction, aimed at addressing information fragmentation among stakeholders. The platform enhanced transparency, traceability and real-time coordination, improving information sharing and project management throughout the modular integrated construction lifecycle. The high computational cost and real-time challenges of blockchain require further research for large-scale implementation. Ensuring a balance between real-time performance and scalability, especially in handling large-scale projects, remains a key issue.

3.2.4. Critical analysis and summary

DT applications in construction have shown considerable promise in enhancing structural monitoring, human-robot collaboration, and modular integration. These applications demonstrated the potential of

DT to enhance safety, automation and resource management in complex environments. However, most existing studies relied on high-quality data and controlled scenarios, limiting their scalability to dynamic, multi-stakeholder settings. Challenges such as fragmented data sources, lack of real-time interoperability, and insufficient empirical validation in large-scale projects persist.

3.3. Transportation

The CII journal has published only 3 articles related to DT use in the transportation sector, as outlined in [Table 4](#). [Errandonea et al. \(2023\)](#) proposed an edge intelligence-based system for diagnosing catenary stagger amplitude in railways, enhancing maintenance efficiency and predictive maintenance strategies through real-time data processing and analysis. The study utilised a hybrid approach combining physics-based simulation data with machine learning model training to improve diagnostic accuracy, offering a practical solution for early fault detection in the railway industry. The system was highly dependent on sensor data quality, particularly in harsh environments or when data is missing. Ensuring system accuracy and robustness under such conditions requires further research and improvement. [Rahman et al. \(2023\)](#) proposed a 3D railway track reconstruction method based on robotic vision, utilising a mobile manipulator and multi-sensor fusion technology to generate high-precision digital models of track components. By integrating global geolocation and laser scanning, the system captures detailed data on track structure and surface conditions, offering intelligent automated support for railway maintenance, including inspection and repair. While the study demonstrated excellent performance in 3D track reconstruction and automated inspection, its adaptability and stability in complex environments, such as harsh weather or low-light conditions, have not been thoroughly explored. Additionally, the system primarily focused on the geometric reconstruction of tracks, lacking comprehensive monitoring of track fatigue and structural health, which limits its potential for broader railway maintenance applications. [Liu et al. \(2023b\)](#) introduced a Cognitive Digital Twin (CDT) framework with enhanced semantic capabilities to address freight parking management in last-mile delivery. The study employed a four-layer architecture that integrates logistics entities and systems into the semantic environment of smart cities. Utilising ontology technology and Web of Things standards, the framework optimised logistics operations planning and urban resource allocation. A real-world case study conducted in Paris demonstrated the potential of CDT in improving logistics efficiency, reducing traffic externalities, and better utilising urban resources. Handling large volumes of heterogeneous data while ensuring the real-time consistency and optimisation of semantic processing and decision-making remains a technical challenge, particularly in large-scale logistics scenarios.

DT applications in transportation have shown considerable promise in predictive maintenance, intelligent inspection, and logistics coordination. These efforts highlighted the potential to improve operational efficiency and asset reliability. However, the studies relied on ideal data conditions and limited-scale validations, reducing their robustness in real-world environments. Challenges remain in handling noisy or sparse data, ensuring interoperability and adapting to complex, dynamic transport systems.

Table 4
Summary of DT applications in transportation industry as published in CII journal.

Specific Focus	Refs.
Predictive maintenance in the railway sector	(Errandonea et al., 2023)
Railway track reconstruction	(Rahman et al., 2023)
Freight parking management	(Liu et al., 2023b)

3.4. Technologies and paradigms

Technologies and paradigms need to continuously evolve in their applications to adapt to increasing demands. The CII journal has published 23 articles on technologies and paradigms involving DTs. These articles can be categorised into three main groups, namely frameworks, architectures and interoperability, intelligence and autonomy, and security, maintenance and lifecycle optimisation, as presented in Table 5.

3.4.1. Frameworks, architectures and interoperability

Studies on DT frameworks, architectures, and interoperability emphasise advancements in modularity, semantic consistency and integration. Boyes and Watson (2022) introduced a comprehensive framework to standardise the analysis and comparison of DT, addressing inconsistencies in current definitions and laying out future research directions for improving DT interoperability and functionality. OMILAB is a Digital Innovation environment that uses agile conceptual modelling to bridge business needs with digital and physical twins for efficient PSS development (Karagiannis et al., 2022). OpenTwins provides an open-source framework for creating compositional DTs, enabling seamless integration of IoT, 3D visualisation, and real-time machine learning for enhanced decision-making (Robles et al., 2023). The practical application and industry adoption of the DT standardisation framework have yet to be validated. Additionally, there is a lack of

Table 5
Summary of DT applications in technologies and paradigms.

Categorisation	Specific Focus	Refs.
Frameworks, Architectures and Interoperability	Standardised analysis framework	(Boyes and Watson, 2022)
	Digital innovation environment	(Karagiannis et al., 2022)
	Open-source DT framework	(Robles et al., 2023)
	Semantic interoperability using AAS	(Rongen et al., 2023)
	Hierarchical architecture for smart factories	(Lippi et al., 2023)
	Multiverse paradigm to harmonise and integrate interpretations	(Cimino et al., 2021)
	Modular platform for cost-efficient development	(Panarotto et al., 2023)
	Aggregated DT integrated with services	(Human et al., 2023)
	Long-term sustainable management	(Zeb et al., 2022)
	Value creation in DT products	(Medina and Hernandez, 2025)
Intelligence and Autonomy	Adaptive DT with microservices	(Bellavista et al., 2023)
	Neural networks and fuzzy logic	(Rodríguez et al., 2023)
	Fault detection in rotating machinery	(Hu et al., 2023a)
	BDI agent-based twins for decision-making	(Alelaimat et al., 2023)
	Cognitive twins with AI governance	(Kalaboukas et al., 2023)
	DT testbed: Simplexity	(Wang et al., 2023)
	Testbed	
Security, Maintenance and Lifecycle Optimisation	XR and DES integration	(Scurati et al., 2025)
	Classification framework for DT use cases	(Abisset-Chavanne et al., 2024)
	Best practices for communication in DT projects	(Kober et al., 2024)
	3D semantic and instance segmentation	(Birkeland and Udnæs, 2024)
	Security management	(Suhail et al., 2023)
	Blockchain for DT security	(Suhail et al., 2022)
	Literature review of predictive maintenance	(Errandonea et al., 2020)

in-depth exploration in areas such as large-scale complex systems, data management, and cross-platform data integration. Rongen et al. (2023) proposed a hybrid approach using Asset Administration Shell (AAS) and Resource Description Framework models to enhance semantic interoperability in DTs, offering improved modularity and flexibility for industrial applications. Lippi et al. (2023) proposed a hierarchical DT architecture for smart factories, employing causal models to improve the system's ability to adapt to evolving operational scenarios through data-driven decision-making.

Cimino et al. (2021) introduced the concept of a DT multiverse, which provides a higher level of abstraction to harmonise diverse interpretations of DT across disciplines, thereby enhancing consistency and facilitating knowledge sharing in industrial systems. Research on semantic interoperability and modular design has enhanced the flexibility and interoperability of DT. However, scalability, data consistency, and practical operability in cross-industry applications and complex multi-system integration still require further validation and optimisation. Panarotto et al. (2023) presented a modular platform approach for the cost-efficient development of DT, focusing on maximising the reuse of physical testing data to validate new design concepts in advanced systems such as aerospace. Human et al. (2023) introduced a framework for designing aggregated DTs, focusing on scalability, modularity and multi-stakeholder environments, with case studies in smart city and heliostat applications. Zeb et al. (2022) addressed strategies for sustaining long-term functional DTs, emphasising the use of open standards and maintaining original infrastructure to ensure data preservation and adaptability. Medina and Hernandez (2025) reviewed the existing literature on product DTs, highlighting value creation mechanisms and identifying gaps. They proposed a five-point research agenda to deepen understanding of DT organisational benefits. These studies have shown the flexibility of modular design and intelligent governance in domains like aviation, smart cities and supply chain management. However, challenges persist in handling large-scale system complexity, stakeholder coordination and real-time data governance. The lack of empirical research and comprehensive solutions is especially evident when dealing with multi-level uncertainty and dynamic changes in these scenarios.

3.4.2. Intelligence and autonomy

DT technology enhances intelligence and autonomy through adaptive design, real-time monitoring and automated decision-making. Bellavista et al. (2023) outlined the design requirements and patterns for adaptive DTs in Industry 4.0, focusing on microservices to enable flexibility, autonomy and context-awareness in digital factory environments. Rodríguez et al. (2023) introduced a novel methodology combining neural networks and fuzzy logic to maintain the accuracy of DT, ensuring reliable real-time data flow between physical and digital spaces for enhanced industrial control. Hu et al. (2023a) proposed a novel Wasserstein generative DT model for health monitoring and fault detection in rotating machines, offering high fidelity and autonomous diagnosis without requiring prior knowledge of faults. Alelaimat et al. (2023) provided an automated toolkit for generating Belief-Desire-Intention (BDI) agent-based DTs, presenting an abductive reasoning approach to model organisational behaviour and improve real-time decision-making processes. Kalaboukas et al. (2023) presented a governance framework for cognitive DTs in supply chains, emphasising sustainability, data governance and AI model management to ensure adaptive, autonomous operations and improved resilience to disruptions. Simplexity Testbed provides a flexible DT environment for testing autonomous systems, integrating multiple modelling paradigms to simulate complex scenarios with high uncertainty (Wang et al., 2023). Scurati et al. (2025) reviewed the integration of Extended Reality (XR) with Discrete Event Simulations (DES) to enhance DT applications. They explored design trade-offs, best practices and used a case study in the mining sector. The study illustrated the benefits of combining XR and DES for more immersive and effective simulations. DT technology has

shown potential in health monitoring, fault detection, and automated decision-making. However, challenges remain in real-time data processing, multi-source data integration as well as ensuring system flexibility and consistency in complex and dynamic industrial environments. Additionally, while the integration of DT with AI enhances automation, balancing computational complexity with real-time responsiveness and ensuring model interpretability requires further investigation.

3.4.3. Security, maintenance and lifecycle optimisation

The DT technology enhances security, maintenance and lifecycle optimisation by leveraging AI-driven cybersecurity, predictive maintenance and decision support systems. [Abisset-Chavanne et al. \(2024\)](#) proposed a comprehensive framework to classify DT use cases, offering a pragmatic approach for defining and adopting DT technology across multiple industrial sectors. [Kober et al. \(2024\)](#) identified critical communication challenges in DT projects, proposing best practices for enhancing collaboration and communication among stakeholders, particularly in multidisciplinary environments within Industry 4.0 and Industry 5.0. [Birkeland and Udnæs \(2024\)](#) proposed a semi-automated method to generate labelled datasets for industrial point clouds, improving the efficiency of creating geometric DTs for complex industrial settings. [Suhail et al. \(2023\)](#) introduced ENIGMA, an AI-driven security solution for DT use in cyber-physical systems, which combines eXplainable AI and gamification to enhance cybersecurity training for analysts. The framework leverages gamified attack-defence scenarios, making AI-driven decisions more transparent and trustworthy through the use of Shapley Additive Explanations values. [Suhail et al. \(2022\)](#) proposed a blockchain-based framework for situational awareness in cyber-physical systems, leveraging DT technology to detect anomalies and improve system security throughout the lifecycle. [Errandonea et al. \(2020\)](#) reviewed DT applications in maintenance, highlighting trends and challenges in predictive and condition-based maintenance, with a focus on decision support and lifecycle management. The varying demands and definitions of DT across different fields make it challenging to achieve standardised interoperability. Although existing research has proposed standardisation frameworks and classification methods for DTs, their universality and consistency in cross-industry applications remain insufficiently validated. Maintaining consistency in data formats, semantic models and technical architectures across diverse domains, while maintaining flexibility and

scalability in cross-industry environments, continues to be a key obstacle to the widespread adoption of DT.

3.4.4. Critical analysis and summary

DT applications in system architectures, autonomous intelligence, and lifecycle management have shown considerable promise in enhancing decision-making and system coordination. While these frameworks lay a strong theoretical foundation, most remain untested at scale. Key challenges include high computational demand, cross-platform integration difficulties, and limited explainability of AI-driven components, all of which hinder practical deployment.

3.5. Practical perspectives in major companies

A review of articles published in CII highlights substantial advancements in DT research within manufacturing, construction, and transportation, offering theoretical and technical foundations for industry applications. In 2023, the DT market was valued at USD 16.75 billion, with a projected compound annual growth rate of 35.7 % from 2024 to 2030 ([Khanal and Shah, 2024](#)). Rising market demand has driven leading companies to adopt DT technology in practice. [Fig. 9](#) depicts DT diverse applications across leading firms. For instance, in 2012, General Motors started using remote monitoring and fault diagnosis for wind turbines. Moreover, Siemens, Dassault and ABB have expanded DT use in reducing research and development cycles and improving product quality with techniques like 3D modelling, equipment monitoring and virtual testing. Honeywell's DT model streamlines aero-engine maintenance and predicts engine performance. United Parcel Service implemented DT technology with real-time analytics to manage its complex logistics. Other major companies such as Microsoft, Amazon, Rockwell Automation, ANSYS, Autodesk, AVEVA, Bentley Systems, General Electric, Hexagon, Parametric Technology Corporation, Bosch and SAP hold significant share in the DT market. Despite the advances of DT applications in these major companies, standardising DT technologies to ensure global interoperability is still a challenging task.

Dedicated businesses in various industries (i.e., manufacturing, construction and transportation) offer specialised and tailored DT solutions. For instance, Fabrik, an Indian startup, provided an operational DT solution that streamlines live manufacturing data and captures plant errors in real-time. Miura Simulation, a French startup, provided real-

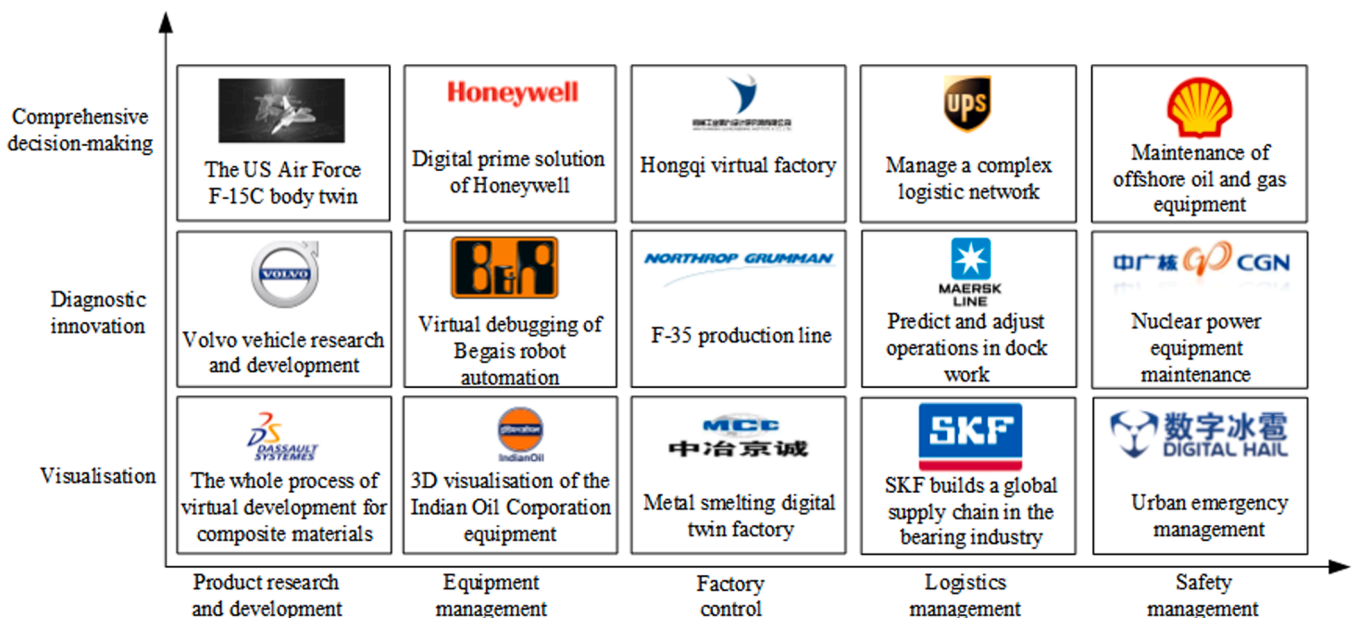


Fig. 9. An application map of DT in major companies.

time simulation twins for aircraft manufacturing, enabling operators to predict behaviour and quality, thereby helping manufacturers reduce costs, save time and gain insights into material performance. Quick AI-algorithm, an Italian startup, has developed an AI-based computerised maintenance management system. This system aggregates manufacturing data to develop a DT, enhancing plant management. It also enables anomaly identification, energy consumption analysis and cost reduction by integrating IoT-based solutions. Sensat, a UK-based startup, developed a DT for the construction industry to enhance project management and infrastructure projects. Swarm Logistics, a German startup, offered Auto-Dispatcher, which is an AI-powered DT solution that optimises dispatching, planning and scheduling for diverse transport fleets.

Startups dedicated to other industries (i.e., energy, healthcare, agriculture, finance, oil & gas and environment & climate) are also thriving. SEKAI, a Swiss startup, developed 'Spin Doctor', which is a DT software program for wind turbines that uses phantom sensors to collect comprehensive turbine data and a proprietary knowledge graph, aiding wind farm managers in predictive maintenance and sensor outage detection. Verto Health, a Canadian startup, proposed a DT orchestration platform that integrates with healthcare systems to centralise patient and process data across clinics, labs and hospitals. Agrimetrics, a UK-based company, was dedicated to integrating DT technology into agriculture by providing a data marketplace for the food and farming sector. Their DT assists in managing natural capital, tracking environmental changes and improving overall farm sustainability. TODAQ, another Canadian startup, facilitated peer-to-peer trading through decentralised DT agents that integrate identity management into finance software systems, enabling businesses to incorporate digital trade in a more cost-effective way. Dinur Soft, a UK-based startup, enhanced oil and gas well management using DT technology, machine learning and predictive analytics to detect anomalies. ClimaTwin, a US-based startup, offered climate risk intelligence for infrastructure development to identify environmental vulnerabilities. Athena SmartCities, another US startup, is dedicated to transforming urban areas into more efficient and sustainable communities.

4. Research opportunities

In this section, we identify and discuss some research opportunities regarding the application of DTs in the industry. This discussion is organised along four broad categories: lifecycle management, cross-disciplinary integration, human-machine collaboration and DT-driven smart industries.

4.1. Extending life-cycle management

DT technology enables dynamic, real-time interaction between the virtual and physical worlds, providing a continuous stream of data that offers valuable insights into the physical asset's performance throughout its lifecycle. By dynamically adapting to the evolving physical asset, DTs support long-term sustainability, continuously reflecting changes. Given the substantial initial investment required to implement DTs, their application should extend across the entire lifecycle to maximise sustainability and optimise long-term resource utilisation. Fig. 10 illustrates the managerial aspects of DT across four main stages (i.e., design, manufacture, operation and disposal) of the PLM (Hribernik et al., 2021; Woitsch et al., 2022).

In most existing studies, the application of DTs has primarily focused on a single function or phase, such as design and innovation, where it is used to test and adjust parameters in a virtual environment. To fully realise the potential of DTs, a key research opportunity lies in broadening its functionalities and extending its application across the entire product lifecycle (Melesse et al., 2020).

Table 6 presents key publications to demonstrate how a DT is applied at different stages of lifecycle management. The research opportunity

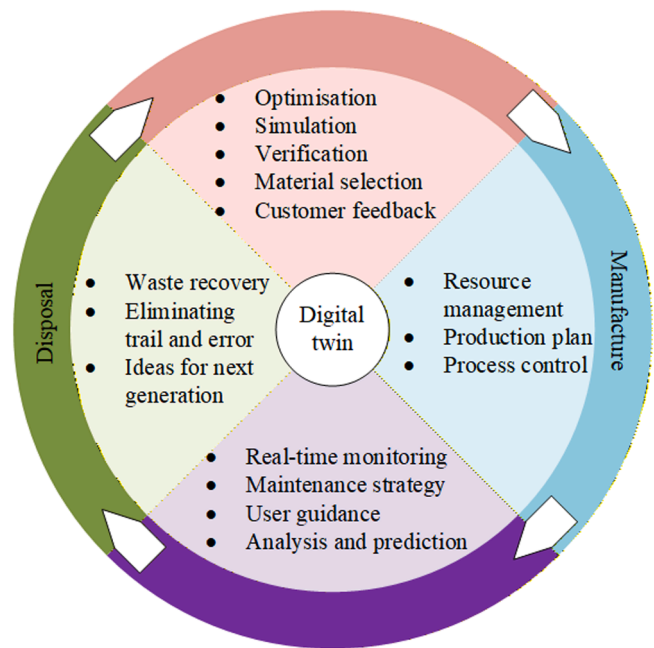


Fig. 10. Four stages of DT throughout the product lifecycle.

we identify is grounded in an extensive review of prestigious journals, including CII, *Advanced Engineering Informatics*, the *International Journal of Production Economics*, and others. Notably, CII has published a wide range of studies that cover all four stages of the PLM, demonstrating the journal's comprehensive scope and depth in this area. Other leading journals have also contributed valuable insights to each stage of the lifecycle, which strengthens the rigour and credibility of the research opportunities identified here. In the design stage, the DT primarily facilitates cost-effective design in a virtual environment. In the manufacturing stage, the DT primarily optimises resource management and production processes to ensure efficient production and reduce production costs. In the operational stage, the DT is predominantly used to enhance quality and reliability through real-time monitoring, data analysis and predictive maintenance, while supporting remote diagnostics and continuous optimisation based on user feedback. The majority of studies have typically focused on only one aspect of lifecycle management. Once the primary functionality is addressed, the DT can be expanded to integrate other highly-coupled functions, progressively moving towards achieving comprehensive full lifecycle management (Lombardo et al., 2024).

4.2. Cross-disciplinary integration

Advances in computer science and technology have improved the capabilities of DT. By integrating various advanced technologies, DTs support a range of applications that enhance system performance and functionality. By evaluating related research published in CII, we have identified a research opportunity related to DT technological innovation and cross-disciplinary integration. With similar studies appearing in other prestigious journals, this opportunity strongly aligns with the current research needs and trends within the industry.

Table 7 outlines the interdisciplinary fusion of DT with technologies such as machine learning, reinforcement learning, and blockchain. In a DT-integrated system, various advanced technologies play a crucial role in enhancing functionality and performance. AI (e.g., machine learning, reinforcement learning, neural network) improves system adaptability, enabling real-time prediction and intelligent decision-making. Blockchain safeguards the security, transparency, and integrity of sensitive data. Data-driven technologies (e.g., IoT, big data, cloud computing, and

Table 6

DT research topics in product full lifecycle management.

Functions	Applications	Journals	Refs.
Design	Requirements and design patterns	CII	(Bellavista et al., 2023)
	Reutilisation and recomposition for variant creation	CII	(Panarotto et al., 2023)
	Adaptive and autonomous system design for digital factories	CII	(Hribernik et al., 2021)
	New product design	Advanced Engineering Informatics	(Haynes and Yang, 2023)
Manufacture	Energy monitoring for sustainable manufacturing	CII	(Abdoune et al., 2023)
	Process planning and scheduling	CII	(Müller-Zhang et al., 2023)
	Real-time quality assurance for zero-defect manufacturing	CII	(Caiazzo et al., 2022)
	Flexibility-driven manufacturing	International Journal of Production Economics	(Neto et al., 2023)
			(Hu et al., 2023a)
Operation	Health monitoring and diagnosis of rotating machines	CII	(Modoni et al., 2022)
	Monitoring the quality of micro injection moulding processes	CII	(Pires et al., 2023)
	Trust-based decision support for efficient scenario recommendations	CII	
	Oil pipeline risk estimation	Journal of Industrial Information Integration	(Priyanka et al., 2022)
			(Sengupta and Dreyer, 2023)
Disposal	Zero-waste value chains for food	CII	(Wang and Wang, 2019)
	Electrical and electronic equipment recovery	International Journal of Production Research	

edge computing) seamlessly integrate and process large volumes of real-time data, providing precise insights to optimise and automate business processes. Communication technology (e.g., 6 G) primarily enables high-speed, low-latency data transmission (Xia et al., 2023a). Similarly, 3D technologies (e.g., 3D modelling, rendering, animation, scanning & reconstruction, printing, visualisation & display, engines & development platforms, and interaction & immersion) create high-fidelity, interactive three-dimensional models to enhance system visualisation and analytical capabilities (Liu et al., 2023a).

Cross-disciplinary integration presents a significant research opportunity for advancing and maturing the DT technology. By incorporating diverse technologies from various fields, this integration will enhance the adaptability and functionality of DTs, making them more applicable across different industries. It will also drive the evolution of DTs by optimising performance, improving decision-making, and supporting real-time, dynamic interactions between the physical and virtual worlds. In addition to technological integration, interdisciplinary collaboration among engineering, computer science, and management disciplines is equally important. Each discipline contributes unique expertise—from system-level engineering design and algorithm development to organisational implementation and strategic alignment. Fostering such collaboration can accelerate the development of robust, scalable, and context-aware DT systems that address real-world complexity.

Table 7

Analysis of cross-disciplinary DT technologies.

Technologies	Applications	Journals	Refs.
Machine learning	Catalyst production forecasting	CII	(Perno et al., 2023)
Reinforcement learning	Trustworthy recommendation	CII	(Pires et al., 2023)
Transfer learning & graph convolutional neural networks	Cross-domain fault diagnosis for centrifugal pumps	CII	(Xu et al., 2024b)
Active learning & simulation	Predictive modelling in the sawmill industry	CII	(Chabanet et al., 2021)
Blockchain	Decentralised manufacturing	CII	(Mandolla et al., 2019)
Big data	Monitoring of production lines	CII	(Xu et al., 2024a)
IoT	Outlier detection and monitoring for industrial manufacturing	CII	(Tapia et al., 2024)
Big data	Sustainable smart manufacturing	Applied Energy	(Ma et al., 2022a)
IoT	Diagnosis of heart diseases	IEEE Internet of Things Journal	(Elayan et al., 2021)
Cloud computing/Edge computing	Monitoring the oil & gas objects	Computers & Industrial Engineering	(Knebel et al., 2023)
6 G	Intelligent planning and energy-efficient deployment	IEEE Wireless Communications	(Xia et al., 2023a)
3D GIS	Display actual environment	Computers and Electronics in Agriculture	(Cho et al., 2023)
3D printing	Validation instances	International Journal of Production Research	(Liu et al., 2023a)
Extended Reality (VR/AR/MR)	Visualisation	Computers & Industrial Engineering	(García et al., 2022)

4.3. Human-machine collaboration

In Industry 5.0, human-machine collaboration is expected to proliferate across various application scenarios. Robots can assist humans in executing complex and precise tasks, while DT technology can provide the essential technical foundation for seamless integration between humans and machines. While humans are integral to these systems, they may also be the source of uncertainty. This uncertainty highlights the necessity to understand and incorporate human variability into DT models. The challenge lies in capturing and integrating human behaviours, decisions and actions into DT models to improve system accuracy and flexibility. Human-machine collaboration presents a significant research opportunity within the realm of DT, as it offers a transformative synergy by combining human intuition and creativity with machine intelligence and precision. The inclusion of user-friendly interfaces and interactive features, enhanced by technologies such as AR and VR, can significantly increase user engagement to make the human-machine interaction more intuitive. Additionally, the ethical and responsible use of AI in DT systems is essential to align technological advancements with societal values and human ethics. By leveraging these complementary strengths, human-machine collaboration could unlock the full potential of DT to achieve more innovative applications across various industries.

Human-machine collaboration is becoming a pivotal research direction in the area of DT. To achieve human-centric smart

manufacturing in Industry 5.0, a human-based DT framework enhances the process by integrating human characteristics into a DT's system design (Wang et al., 2024c). Automatic recognition of human behaviours is crucial for enabling natural and intuitive human-computer interactions (Rouast et al., 2021). For example, a DT system for a human-assisted robotic cell applies real-time simulation and optimisation to maintain context-awareness, autonomy, and adaptability to human behaviours (Albini et al., 2023). Furthermore, a deep learning-enhanced DT framework addresses safety and reliability in collaborative manufacturing by detecting the operations of both humans and robots, ensuring the system's effectiveness in dynamic environments (Wang et al., 2024b). To support consistent and accurate physical-virtual interactions in human-machine collaboration, a consistency methodology, including checks, model evolution and validation, has been developed to facilitate dynamic physical-virtual reflection within the system (Ma et al., 2022b). Although designing efficient sensors for soft robotics remains a challenge, DT technology has been successfully applied in scenarios such as virtual assembly lines and unmanned warehouses, demonstrating their potential in enhancing operational efficiency and flexibility in various industrial contexts (Jin et al., 2020).

The emphasis on human-machine collaboration in DT technology lies in recognising humans as integral to the system, not just as users but as active contributors to decision-making and system adaptability. Ensuring that DTs remain human-centred is key to creating systems that capture the complexities of human behaviour while enabling machine-driven optimisation. Additionally, effective communication among stakeholders, including operators, engineers, managers, and external partners, is essential for building cohesive and efficient systems. This collaborative approach aligns all parties, fosters real-time insight sharing, and drives the success and long-term sustainability of DT applications. While human-machine collaboration through DTs enhances efficiency and adaptability, ethical considerations remain insufficiently addressed. The widespread use of human monitoring raises serious concerns around data privacy, including opaque data collection, inadequate consent, and the normalisation of workplace surveillance. Moreover, automation-driven DT systems may accelerate labour displacement and erode human agency in operational decisions. These risks are often underrepresented in current discourse, which tends to prioritise technical gains over socio-ethical implications. A more critical and inclusive approach is needed, where transparency, accountability, and worker rights are embedded as foundational design principles.

4.4. Digital twin-driven smart industries

The DT technology is transforming industries by creating context-aware, autonomous and adaptive systems (Bellavista et al., 2023; Hribnik et al., 2021; Rico et al., 2023). This advancement is reshaping sectors such as manufacturing, construction and transportation, propelling each towards unprecedented levels of efficiency, precision and adaptability. In manufacturing, the DT technology is transforming manufacturing into an adaptive, self-learning ecosystem. By enabling real-time pattern detection, risk prediction and process refinement, DTs shift production from static workflows to dynamic, interconnected networks. This proactive optimisation enhances equipment maintenance and supply chain coordination, allowing for precise resource allocation and agile responses. These capabilities support self-optimising and self-healing systems, bringing the industry closer to fully autonomous factories. The DT technology is also progressively transforming the construction industry by enabling comprehensive optimisation across design, building, operation, and maintenance phases. In design, DTs simulate various scenarios to support informed decision-making. During the building phase, real-time monitoring aids in resource allocation and enhances safety management. Similarly, during operation and maintenance, DTs integrate with sensor data to allow precise condition monitoring and predictive maintenance. This

reduces energy consumption and minimises unexpected repairs. Additionally, DTs provide real-time data to optimise response strategies and improve safety during emergency management. Overall, the DT technology offers a more efficient and sustainable approach to building management.

The DT technology is also fundamentally reshaping transportation by creating dynamic, data-driven environments that bridge the physical and digital realms. In traffic management, DTs allow for real-time modelling of complex traffic flows, enabling systems to adjust routes, predict congestion, and reduce emissions through optimised logistics. For infrastructure, DTs continuously monitor the condition of roads, bridges, and other critical assets, predicting wear and proactively scheduling maintenance to prevent costly failures. In fleet management, DTs provide a comprehensive view of vehicle health, allowing operators to anticipate repairs and minimise operational disruptions. During emergencies, DTs enhance response coordination by simulating incident scenarios and optimising resource deployment. Altogether, the DT technology enables a more resilient, adaptive, and sustainable transportation infrastructure, moving the industry towards intelligent, autonomous operations.

The DT technology is also revolutionising various other sectors, including energy, healthcare, agriculture, finance, oil & gas and environment & climate. In energy, DTs support real-time monitoring and predictive maintenance of infrastructure, optimising power generation and reducing asset downtime. Healthcare applications include patient-specific models that simulate treatment outcomes, enhancing personalised care. In agriculture, DTs monitor soil conditions and simulate crop growth to optimise resource use and predict yields. The finance industry benefits from DTs that model financial systems for risk analysis, market forecasting, and fraud detection. In oil and gas industries, the DT technology improves asset integrity management by predicting maintenance needs and enhancing safety. Lastly, DTs in environment and climate sectors simulate environmental systems to monitor pollution, assess climate impact, and support sustainable urban planning.

DTs are redefining the fabric of cross-industry and supply chain networks by creating a cohesive, data-driven ecosystem that bridges physical and digital realms. Through real-time data synchronisation and predictive intelligence, DT technology dismantles conventional silos between industries and across supply chain stages, enabling a new era of collective decision-making, resource optimisation, and strategic foresight. This interconnected framework allows for a seamless exchange of insights, where performance metrics and operational data from one sector or stage of the supply chain can inform and elevate outcomes throughout the entire network. The DT technology thus fosters a paradigm shift from isolated, reactive workflows to integrated, anticipatory systems capable of adapting dynamically to complex, rapidly changing demands. In enabling this transformation, DTs not only enhance resilience and agility but also position industries and supply chains to innovate collaboratively, co-creating sustainable, future-ready infrastructures that can withstand and evolve alongside global challenges. In summary, DTs have emerged as a transformative technology, revolutionising smart industries such as manufacturing, construction, transportation, agriculture, energy, and finance, with the potential to empower even more sectors in the future. The characteristics of smart industries enable the accurate replication and real-time monitoring of physical systems in virtual environments, with continuous updates that reflect changes in the physical world, allowing for real-time optimisation and predictive adjustments. This bidirectional data flow enhances industry efficiency, resource allocation, and risk management.

5. Future challenges and managerial insights

Numerous publications and practical applications underscore the significance and benefits of DTs. While embracing the potential of DTs, it is also important to understand the key challenges associated with DT applications. DT technology faces several challenges common to many

large-scale software solutions in the industry, such as high initial investment, system development complexities, scalability and flexibility, reliance on highly specialised expertise, user acceptance, and effective communication of outputs and predictions to management. In addition, DTs encounter the following unique challenges:

- **High-Quality, Real-Time Data Collection & Synchronisation:** Ensuring the seamless synchronisation of high-quality, real-time data between the virtual model and the physical world, while minimising latency and discrepancies (Sharma et al., 2022) is a major challenge for DTs.
- **Integration Complexity:** The DT paradigm demands seamless integration with existing systems, such as legacy IT infrastructure, IoT devices, and big data platforms. This process involves addressing data compatibility, infrastructure upgrades, and security, which requires cross-domain collaboration and adds to the system's complexity (Han et al., 2023).
- **Standardisation and Regulation:** The lack of standardised guidelines and regulations for DTs hinders interoperability and broad adoption across industries (Lucchi, 2023).
- **Data Security and Privacy:** Given the high frequency and amount of bi-directional data-flow, safeguarding the continuous flow of data, ensuring robust security measures and privacy protections to prevent data breaches and malicious attacks are major challenges in DT research and development (Min et al., 2019).
- **Cross-Disciplinary Collaboration:** Effective collaboration across diverse technical domains is crucial, as differences in expertise and communication styles may hinder the successful integration of DT technologies (Yang et al., 2023).
- **Bidirectional Real-Time Interaction Between Physical and Virtual Entities:** Achieving precise real-time interaction between physical and virtual entities, where changes in the physical world are accurately reflected in the virtual model, and vice versa, requires very sophisticated data flows and control mechanisms (Ma et al., 2024).
- **DT Stakeholder Communication:** Effectively conveying complex DT outputs and predictions to stakeholders, particularly management, remains a key challenge, requiring clear, actionable insights, transparency and effective visualisation methods to support informed decision-making (Kober et al., 2024).

Based on the challenges associated with the application of DT technology across industries, we propose several management insights. These insights aim to guide businesses and industries in effectively managing and deploying DT technology, thereby fostering its widespread adoption and maximising its value:

- The establishment of open standards and platforms is essential for facilitating the seamless integration of DT systems with physical assets. Industry consortia, such as the Industrial DT Association, promote shared architectures and protocols that help reduce fragmentation, minimise the need for frequent updates, and avoid large-scale overhauls.
- Enhancing the compatibility and security of data access is paramount to ensuring effective cross-industry data sharing and safeguarding privacy.
- Broadening the application of DT technology and fostering cross-industry collaboration will expedite its adoption and promote cross-sector integration.
- Extending the operational capabilities of DT systems, including real-time monitoring, predictive maintenance, and process optimisation, will further enhance efficiency and effectiveness across various industries.
- The development of supporting policy frameworks, along with the promotion of talent development and technical expertise, will provide a robust foundation for the sustainable advancement and long-term innovation of DT technology.

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CRediT authorship contribution statement

Sai-Ho Chung: Writing – review & editing, Supervision, Funding acquisition. **Hoi-Lam Ma:** Writing – review & editing, Supervision, Funding acquisition. **Debiao Li:** Writing – review & editing, Investigation, Funding acquisition. **Dewang Chen:** Writing – review & editing, Investigation, Funding acquisition. **Sandeep Dhakal:** Writing – review & editing, Validation, Methodology, Investigation. **Raymond Chiong:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Shi Qiang Liu:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Peipei Ding:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Abdoune, F., Ragazzini, L., Nouri, M., Negri, E., Cardin, O., 2023. Toward digital twin for sustainable manufacturing: a data-driven approach for energy consumption behavior model generation. *Comput. Ind.* 150 (May), 103949. <https://doi.org/10.1016/j.compind.2023.103949>.
- Abisset-Chavanne, E., Coupaye, T., Golra, F.R., Lamy, D., Piel, A., Scart, O., Vicat-Blanc, P., 2024. A digital twin use cases classification and definition framework based on industrial feedback. *Comput. Ind.* 161 (October), 104113. <https://doi.org/10.1016/j.compind.2024.104113>.
- Albini, T., Brocchi, A., Murgia, G., Pranzo, M., 2023. Real-time optimisation for a digital twin of a robotic cell with human operators. *Comput. Ind.* 146 (January), 103858. <https://doi.org/10.1016/j.compind.2023.103858>.
- Alalaimat, A., Ghose, A., Dam, H.K., 2023. XPLaM: a toolkit for automating the acquisition of BDI agent-based digital twins of organizations. *Comput. Ind.* 145 (December 2022), 103805. <https://doi.org/10.1016/j.compind.2022.103805>.
- AS, D.S., Antony, A.J., 2017. Experimental study on digital twin spark ignition gasoline engine at different gasoline-methanol blends. *Int. J. Appl. Eng. Res.* 12 (13), 3817–3821.
- Ashraf, M.M., Hasan, N., Lewis, L., Hasan, M.R., Ray, P., 2016. A systematic literature review of the application of information communication technology for visually impaired people. *Int. J. Disabil. Manag.* 11, 1–18. <https://doi.org/10.1017/idm.2016.6>.
- Beerepoot, I., Di Ciccio, C., Reijers, H.A., Rinderle-Ma, S., Bandara, W., Burattin, A., Calvanese, D., Chen, T., Cohen, I., Depaire, B., Di Federico, G., Dumas, M., van Dun, C., Fehrer, T., Fischer, D.A., Gal, A., Indulska, M., Isahagian, V., Klinkmüller, C., Zerbato, F., 2023. The biggest business process management problems to solve before we die. *Comput. Ind.* 146 (January), 103837. <https://doi.org/10.1016/j.compind.2022.103837>.
- Bellavista, P., Biccocchi, N., Fogli, M., Giannelli, C., Mamei, M., Picone, M., 2023. Requirements and design patterns for adaptive, autonomous, and context-aware digital twins in industry 4.0 digital factories. *Comput. Ind.* 149 (April). <https://doi.org/10.1016/j.compind.2023.103918>.
- Bertoni, M., Bertoni, A., 2022. Designing solutions with the product-service systems digital twin: what is now and what is next? *Comput. Ind.* 138, 103629. <https://doi.org/10.1016/j.compind.2022.103629>.
- Bibri, S.E., Huang, J., Jagatheesaperumal, S.K., Krogstie, J., 2024. The synergistic interplay of artificial intelligence and digital twin in environmentally planning sustainable smart cities: a comprehensive systematic review. *Environ. Sci. Ecotechnol.* 20, 100433. <https://doi.org/10.1016/j.es.2024.100433>.

- Birkeland, A.A., Udnæs, M., 2024. Semi-automated dataset creation for semantic and instance segmentation of industrial point clouds. *Comput. Ind.* 155 (December). <https://doi.org/10.1016/j.compind.2023.104064>.
- Borangi, T., Trentesaux, D., Thomas, A., Leitão, P., Barata, J., 2019. Digital transformation of manufacturing through cloud services and resource virtualization. *Comput. Ind.* 108, 150–162. <https://doi.org/10.1016/j.compind.2019.01.006>.
- Böttger, T., Tola, D., Kakavandi, F., Wewer, C.R., Ramanujan, D., Gomes, C., Larsen, P.G., Iosifidis, A., 2023. A review of unit level digital twin applications in the manufacturing industry. In: *CIRP Journal of Manufacturing Science and Technology*, 45. Elsevier Ltd, pp. 162–189. <https://doi.org/10.1016/j.cirpj.2023.06.011>.
- Boyes, H., Watson, T., 2022. Digital twins: an analysis framework and open issues. *Comput. Ind.* 143 (June), 103763. <https://doi.org/10.1016/j.compind.2022.103763>.
- Brereton, P., Kitchenham, B.A., Budgen, D., Turner, M., Khalil, M., 2007. Lessons from applying the systematic literature review process within the software engineering domain. *J. Syst. Softw.* 80 (4), 571–583. <https://doi.org/10.1016/j.jss.2006.07.009>.
- Caiazza, B., Di Nardo, M., Murino, T., Petrillo, A., Piccirillo, G., Santini, S., 2022. Towards zero defect manufacturing paradigm: a review of the state-of-the-art methods and open challenges. *Comput. Ind.* 134, 103548. <https://doi.org/10.1016/j.compind.2021.103548>.
- Calvo-Bascones, P., Voisin, A., Do, P., Sanz-Bobi, M.A., 2023. A collaborative network of digital twins for anomaly detection applications of complex systems. *Snitch digital twin concept. Comput. Ind.* 144 (September), 103767. <https://doi.org/10.1016/j.compind.2022.103767>.
- Cameron, D.B., Waaler, A., Komulainen, T.M., 2018. Oil and gas digital twins after twenty years. How can they be made sustainable, maintainable and useful? In: *Proceedings of the 59th Conference on Imulation and Modelling (SIMS 59)*, 26–28 September 2018, 153 Oslo Metropolitan University, Norway, pp. 9–16. <https://doi.org/10.3384/ecp181539>.
- Chabanet, S., Bril El-Haouzi, H., Thomas, P., 2021. Coupling digital simulation and machine learning metamodel through an active learning approach in industry 4.0 context. *Comput. Ind.* 133, 103529. <https://doi.org/10.1016/j.compind.2021.103529>.
- Chatterjee, P., Das, D., Rawat, D.B., 2024. Digital twin for credit card fraud detection: opportunities, challenges, and fraud detection advancements. *Future Gener. Comput. Syst.* 158 (April), 410–426. <https://doi.org/10.1016/j.future.2024.04.057>.
- Cho, J., Kim, C., Lim, K.J., Kim, J., Ji, B., Yeon, J., 2023. Web-based agricultural infrastructure digital twin system integrated with GIS and BIM concepts. *Comput. Electron. Agric.* 215 (July), 108441. <https://doi.org/10.1016/j.compag.2023.108441>.
- Christou, I.T., Kefalakis, N., Soldatos, J.K., Despotopoulou, A.M., 2022. End-to-end industrial IoT platform for quality 4.0 applications. *Comput. Ind.* 137, 103591. <https://doi.org/10.1016/j.compind.2021.103591>.
- Cimino, C., Ferretti, G., Leva, A., 2021. Harmonising and integrating the digital twins multiverse: a paradigm and a toolset proposal. *Comput. Ind.* 132, 103501. <https://doi.org/10.1016/j.compind.2021.103501>.
- Cimino, C., Negri, E., Fumagalli, L., 2019. Review of digital twin applications in manufacturing. *Comput. Ind.* 113, 103130. <https://doi.org/10.1016/j.compind.2019.103130>.
- Dattatreya, Z., Gaurav, S., Bhagyashri, P., 2014. Digital twin spark ignition using mechatronics. *IOSR J. Mech. Civ. Eng.* 2014, 73–78. (<https://www.academia.edu/download/81819971/16.pdf>).
- De Beer, J., Engels, J., Heerkens, Y., Van Der Klink, J., 2014. Factors influencing work participation of adults with developmental dyslexia: a systematic review. *BMC Public Health* 14 (1). <https://doi.org/10.1186/1471-2458-14-77>.
- De Giacomo, G., Favorito, M., Leotta, F., Mecella, M., Silo, L., 2023. Digital twins composition in smart manufacturing via Markov decision processes. *Comput. Ind.* 149 (April), 103916. <https://doi.org/10.1016/j.compind.2023.103916>.
- de Oliveira Hansen, J.P., Ribeiro da Silva, E., Bilberg, A., 2024. Agile digital machine development. *Comput. Ind.* 155 (December), 104061. <https://doi.org/10.1016/j.compind.2023.104061>.
- Dieste, O., Grímán, A., Juristo, N., 2009. Developing search strategies for detecting relevant experiments. *Empir. Softw. Eng.* 14 (5), 513–539. <https://doi.org/10.1007/s10664-008-9091-7>.
- Ebrahimighahnavieh, M.A., Luo, S., Chiong, R., 2020. Deep learning to detect Alzheimer's disease from neuroimaging: a systematic literature review. *Comput. Methods Prog. Biomed.* 187. <https://doi.org/10.1016/j.cmpb.2019.105242>.
- Elayan, H., Aloqaily, M., Guizani, M., 2021. Digital twin for intelligent Context-Aware IoT healthcare systems. *IEEE Internet Things J.* 8 (23), 16749–16757. <https://doi.org/10.1109/JIOT.2021.3051158>.
- Errandonea, I., Beltrán, S., Arrizabalaga, S., 2020. Digital twin for maintenance: a literature review. *Comput. Ind.* 123. <https://doi.org/10.1016/j.compind.2020.103316>.
- Errandonea, I., Ciáurriz, P., Alvarado, U., Beltrán, S., Arrizabalaga, S., 2023. Edge intelligence-based proposal for onboard catenary stagger amplitude diagnosis. *Comput. Ind.* 144 (June), 1–12. <https://doi.org/10.1016/j.compind.2022.103781>.
- Fawad, M., Salamak, M., Chen, Q., Uscilowski, M., Koris, K., Jasinski, M., Lazinski, P., Piotrowski, D., 2025. Development of immersive bridge digital twin platform to facilitate bridge damage assessment and asset model updates. *Comput. Ind.* 164 (April), 104189. <https://doi.org/10.1016/j.compind.2024.104189>.
- Friederich, J., Francis, D.P., Lazarova-Molnar, S., Mohamed, N., 2022. A framework for data-driven digital twins for smart manufacturing. *Comput. Ind.* 136, 103586. <https://doi.org/10.1016/j.compind.2021.103586>.
- Gaffinet, B., Al Haj Ali, J., Naudet, Y., Panetto, H., 2025. Human digital twins: a systematic literature review and concept disambiguation for industry 5.0. *Comput. Ind.* 166 (July), 104230. <https://doi.org/10.1016/j.compind.2024.104230>.
- Garcia, F.A., Devriendt, H., Metin, H., Özer, M., Naets, F., 2025. Physics-informed digital twin design for supporting the selection of process settings in continuous manufacturing, with a focus in fiberboard production. *Comput. Ind.* 168 (February), 104267. <https://doi.org/10.1016/j.compind.2025.104267>.
- García, Á., Bregon, A., Martínez-Prieto, M.A., 2022. Towards a connected digital twin learning ecosystem in manufacturing: enablers and challenges. *Comput. Ind. Eng.* 171 (July), 108463. <https://doi.org/10.1016/j.cie.2022.108463>.
- Gregory, C., Mbolamananamalala, R., Rabah, S., Chapurlat, V., 2024. Model based systems engineering applied to digital twin engineering: why and how to? *IFACPaper* 58 (19), 157–162. <https://doi.org/10.1016/j.ifacol.2024.09.126>.
- Greif, T., Stein, N., Flath, C.M., 2020. Peeking into the void: digital twins for construction site logistics. *Comput. Ind.* 121, 103264. <https://doi.org/10.1016/j.compind.2020.103264>.
- Grievies M. 2014. Digital twin: manufacturing excellence through virtual factory replication. In *White paper* (Vol. 1, Issue March).
- Grievies, M., Vickers, J., 2016. Digital twin: mitigating unpredictable, undesirable emergent behavior in complex systems. *Transdiscipl. Perspect. Complex Syst. N. Find. Approach.* 85–113. https://doi.org/10.1007/978-3-319-38756-7_4.
- Han, Y., Li, Y., Li, Y., Yang, B., Cao, L., 2023. Digital twinning for smart hospital operations: framework and proof of concept. *Technol. Soc.* 74. <https://doi.org/10.1016/j.techsoc.2023.102317>.
- Haynes, P., Yang, S., 2023. Supersystem digital twin-driven framework for new product conceptual design. *Adv. Eng. Inform.* 58. <https://doi.org/10.1016/j.aei.2023.102149>.
- Hribernik, K., Cabri, G., Mandreoli, F., Mentzas, G., 2021. Autonomous, context-aware, adaptive digital Twins—State of the art and roadmap. *Comput. Ind.* 133, 103508. <https://doi.org/10.1016/j.compind.2021.103508>.
- Hu, W., Wang, T., Chu, F., 2023a. A wasserstein generative digital twin model in health monitoring of rotating machines. *Comput. Ind.* 145 (April), 103807. <https://doi.org/10.1016/j.compind.2022.103807>.
- Hu, F., Wang, W., Zhou, J., 2023b. Petri nets-based digital twin drives dual-arm cooperative manipulation. *Comput. Ind.* 147 (January), 103880. <https://doi.org/10.1016/j.compind.2023.103880>.
- Human, C., Basson, A.H., Kruger, K., 2023. A design framework for a system of digital twins and services. *Comput. Ind.* 144 (June), 103796. <https://doi.org/10.1016/j.compind.2022.103796>.
- Isaja, M., Nguyen, P., Goknil, A., Sen, S., Husom, E.J., Tverdal, S., Anand, A., Jiang, Y., Pedersen, K.J., Myrseth, P., Stang, J., Njavi, H., Pfeiffer, S., Lamplair, P., 2023. A blockchain-based framework for trusted quality data sharing towards zero-defect manufacturing. *Comput. Ind.* 146 (November), 103853. <https://doi.org/10.1016/j.compind.2023.103853>.
- Jacoby, M., Uslander, T., 2020. Digital twin and Internet of things-Current standards landscape. *Appl. Sci.* 10 (18), 1–21. <https://doi.org/10.3390/AP10186519>.
- Jbair, M., Ahmad, B., Maple, C., Harrison, R., 2022. Threat modelling for industrial cyber physical systems in the era of smart manufacturing. *Comput. Ind.* 137, 103611. <https://doi.org/10.1016/j.compind.2022.103611>.
- Jiang, Y., Li, M., Guo, D., Wu, W., Zhong, R.Y., Huang, G.Q., 2022. Digital twin-enabled smart modular integrated construction system for on-site assembly. *Comput. Ind.* 136, 103594. <https://doi.org/10.1016/j.compind.2021.103594>.
- Jiang, Y., Liu, X., Kang, K., Wang, Z., Zhong, R.Y., Huang, G.Q., 2021. Blockchain-enabled cyber-physical smart modular integrated construction. *Comput. Ind.* 133, 103553. <https://doi.org/10.1016/j.compind.2021.103553>.
- Jin, T., Sun, Z., Li, L., Zhang, Q., Zhu, M., Zhang, Z., Yuan, G., Chen, T., Tian, Y., Hou, X., Lee, C., 2020. Triboelectric nanogenerator sensors for soft robotics aiming at digital twin applications. *Nat. Commun.* 11 (1). <https://doi.org/10.1038/s41467-020-19059-3>.
- Jo, S.K., Park, D.H., Park, H., Kim, S.H., 2018. Smart livestock farms using digital twin: feasibility study. 9th Int. Conf. Inf. Commun. Technol. Converg. ICT Converg. Power Smart Intell. ICTC 2018 1461–1463. <https://doi.org/10.1109/ICTC.2018.8539516>.
- Jorgensen, J., Hoddiewicz, M., Cripps, E., Hassan, G.M., 2023. Requirements for the application of the digital twin paradigm to offshore wind turbine structures for uncertain fatigue analysis. *Comput. Ind.* 145 (September), 103806. <https://doi.org/10.1016/j.compind.2022.103806>.
- Kalaboukas, K., Kiritsis, D., Arampatzis, G., 2023. Governance framework for autonomous and cognitive digital twins in agile supply chains. *Comput. Ind.* 146 (April), 103857. <https://doi.org/10.1016/j.compind.2023.103857>.
- Karagiannis, D., Buchmann, R.A., Utz, W., 2022. The OMILAB digital innovation environment: agile conceptual models to bridge business value with digital and physical twins for product-service systems development. *Comput. Ind.* 138, 103631. <https://doi.org/10.1016/j.compind.2022.103631>.
- Keele, S., 2007. Guidelines for performing systematic literature reviews in software engineering. *Tech. Report. Ver. 2.3 EBSE Tech. Rep. EBSE 1. Issue January 2007*.
- Khanal, A., Shah, A., 2024. Techno-Economic analysis of hemp production, logistics and processing in the U.S. *Biomass* 4 (1), 164–179. <https://doi.org/10.3390/biomass4010008>.
- Knebel, F.P., Trevisan, R., Nascimento, G.S., do, Abel, M., Wickboldt, J.A., 2023. A study on cloud and edge computing for the implementation of digital twins in the oil & gas industries. *Comput. Ind. Eng.* 182, 109363. <https://doi.org/10.1016/j.cie.2023.109363>.
- Kober, C., Medina, F.G., Benfer, M., Wulfsberg, J.P., Martinez, V., Lanza, G., 2024. Digital twin stakeholder communication: characteristics, challenges, and best practices. *Comput. Ind.* 161 (June), 104135. <https://doi.org/10.1016/j.compind.2024.104135>.
- Korth, B., Schwede, C., Zajac, M., 2018. Simulation-ready digital twin for realtime management of logistics systems. *Proc. 2018 IEEE Int. Conf. Big Data Big Data 2018* 4194–4201. <https://doi.org/10.1109/BigData.2018.8622160>.

- Kritzinger, W., Karner, M., Traar, G., Henjes, J., Sihn, W., 2018. Digital twin in manufacturing: a categorical literature review and classification. *IFACPaper* 51 (11), 1016–1022. <https://doi.org/10.1016/j.ifacol.2018.08.474>.
- Liao, Z., Wei, W., Zhang, L., Gao, Y., Cai, Y., 2025. An augmented reality-enabled digital twin system for reconfigurable soft robots: visualization, simulation and interaction. *Comput. Ind.* 168 (October), 104285. <https://doi.org/10.1016/j.compind.2025.104285>.
- Lippi, M., Martinelli, M., Picone, M., Zambonelli, F., 2023. Enabling causality learning in smart factories with hierarchical digital twins. *Comput. Ind.* 148 (April), 103892. <https://doi.org/10.1016/j.compind.2023.103892>.
- Liu, X., Jiang, Y., Wang, Z., Zhong, R.Y., Cheung, H.H., Huang, G.Q., 2023a. Imstudio: blockchain-enabled secure digital twin platform for service manufacturing. *Int. J. Prod. Res.* 61 (12), 3984–4003. <https://doi.org/10.1080/00207543.2021.2003462>.
- Liu, Y., Pan, S., Folz, P., Ramparany, F., Bolle, S., Ballot, E., Coupaye, T., 2023b. Cognitive digital twins for freight parking management in last mile delivery under smart cities paradigm. *Comput. Ind.* 153 (September). <https://doi.org/10.1016/j.compind.2023.104022>.
- Lombardo, G., Picone, M., Mamei, M., Mordonini, M., Poggi, A., 2024. Digital twin for continual learning in location based services. *Eng. Appl. Artif. Intell.* 127. <https://doi.org/10.1016/j.engappai.2023.107203>.
- Long, W., Bao, Z., Chen, K., Thomas Ng, S., Yahaya Wuni, I., 2024. Developing an integrative framework for digital twin applications in the building construction industry: a systematic literature review. *Adv. Eng. Inform.* 59 (January), 102346. <https://doi.org/10.1016/j.aei.2023.102346>.
- Lucchi, E., 2023. Digital twins for the automation of the heritage construction sector. *Autom. Constr.* 156. <https://doi.org/10.1016/j.autcon.2023.105073>.
- Lugaresi, G., Gangemi, S., Gazzoni, G., Matta, A., 2023. Online validation of digital twins for manufacturing systems. *Comput. Ind.* 150 (May). <https://doi.org/10.1016/j.compind.2023.103942>.
- Lugaresi, G., Matta, A., 2023. Automated digital twin generation of manufacturing systems with complex material flows: graph model completion. *Comput. Ind.* 151 (June), 103977. <https://doi.org/10.1016/j.compind.2023.103977>.
- Ma, S., Ding, W., Liu, Y., Ren, S., Yang, H., 2022a. Digital twin and big data-driven sustainable smart manufacturing based on information management systems for energy-intensive industries. *Appl. Energy* 326 (September), 119986. <https://doi.org/10.1016/j.apenergy.2022.119986>.
- Ma, X., Qi, Q., Tao, F., 2024. An ontology-based data-model coupling approach for digital twin. *Robot. Comput. Integr. Manuf.* 86. <https://doi.org/10.1016/j.rcim.2023.102649>.
- Ma, X., Qi, Q., Cheng, J., Tao, F., 2022b. A consistency method for digital twin model of human-robot collaboration. *J. Manuf. Syst.* 65, 550–563. <https://doi.org/10.1016/j.jmsy.2022.10.012>.
- Mandolla, C., Petruzzelli, A.M., Percoco, G., Urbinati, A., 2019. Building a digital twin for additive manufacturing through the exploitation of blockchain: a case analysis of the aircraft industry. *Comput. Ind.* 109, 134–152. <https://doi.org/10.1016/j.compind.2019.04.011>.
- Mantravadi, S., Srail, J.S., Möller, C., 2023. Application of MES/MOM for industry 4.0 supply chains: a cross-case analysis. *Comput. Ind.* 148 (June). <https://doi.org/10.1016/j.compind.2023.103907>.
- Marquez, A.C., Antonio, J.M.A., Lopez, A.J.G., Carmona, A.D.L.F., 2023. Digital twins in condition-based maintenance apps: a case study for train axle bearings. *Comput. Ind.* 151 (June). <https://doi.org/10.1016/j.compind.2023.103980>.
- Medina, F.G., Hernandez, V.M., 2025. Product digital twins: an umbrella review and research agenda for understanding their value. *Comput. Ind.* 164 (July). <https://doi.org/10.1016/j.compind.2024.104181>.
- Melesse, T.Y., Di Pasquale, V., Riemma, S., 2020. Digital twin models in industrial operations: a systematic literature review. *Procedia Manuf.* 42 (2019), 267–272. <https://doi.org/10.1016/j.promfg.2020.02.084>.
- Min, Q., Lu, Y., Liu, Z., Su, C., Wang, B., 2019. Machine learning based digital twin framework for production optimisation in petrochemical industry. *Int. J. Inf. Manag.* 49, 502–519. <https://doi.org/10.1016/j.ijinfomgt.2019.05.020>.
- Mitsanis, C., Hurst, W., Tekinerdogan, B., 2024. A 3D functional plant modelling framework for agricultural digital twins. *Comput. Electron. Agric.* 218 (February), 108733. <https://doi.org/10.1016/j.compag.2024.108733>.
- Modoni, G.E., Stampone, B., Trotta, G., 2022. Application of the digital twin for in process monitoring of the micro injection moulding process quality. *Comput. Ind.* 135, 103568. <https://doi.org/10.1016/j.compind.2021.103568>.
- Mohammadi, M., Rashidi, M., Yu, Y., Samali, B., 2023. Integration of TLS-derived bridge information modeling (BrIM) with a decision support system (DSS) for digital twinning and asset management of bridge infrastructures. *Comput. Ind.* 147 (January). <https://doi.org/10.1016/j.compind.2023.103881>.
- Müller-Zhang, Z., Kuhn, T., Antonino, P.O., 2023. Towards live decision-making for service-based production: integrated process planning and scheduling with digital twins and Deep-Q-Learning. *Comput. Ind.* 149 (January), 103933. <https://doi.org/10.1016/j.compind.2023.103933>.
- Naplekov, I., Zheleznikov, I., Pashchenko, D., Kobysheva, P., Moskvitina, A., Mustafin, R., Gnutikova, M., Mullagalieva, A., Uzlou, P., 2018. Methods of computational modeling of coronary heart vessels for its digital twin. *MATEC Web Conf.* 172, 1–6. <https://doi.org/10.1051/mateconf/201817201009>.
- Neto, A.A., Ribeiro da Silva, E., Deschamps, F., do Nascimento Junior, L.A., Pinheiro de Lima, E., 2023. Modeling production disorder: procedures for digital twins of flexibility-driven manufacturing systems. *Int. J. Prod. Econ.* 260. <https://doi.org/10.1016/j.ijpe.2023.108846>.
- Ogunsakin, R., Mehandjiev, N., Marin, C.A., 2023. Towards adaptive digital twins architecture. *Comput. Ind.* 149 (January). <https://doi.org/10.1016/j.compind.2023.103920>.
- Panarotto, M., Isaksson, O., Vial, V., 2023. Cost-efficient digital twins for design space exploration: a modular platform approach. *Comput. Ind.* 145 (November), 103813. <https://doi.org/10.1016/j.compind.2022.103813>.
- Pantazis, D., Goodall, P., Pease, S.G., Conway, P., West, A., 2023. Predicting electrical power consumption of end milling using a virtual machining energy toolkit (V.MET). *Comput. Ind.* 150 (May), 103943. <https://doi.org/10.1016/j.compind.2023.103943>.
- Perno, M., Hvam, L., Haug, A., 2023. A machine learning digital twin approach for critical process parameter prediction in a catalyst manufacturing line. *Comput. Ind.* 151 (May), 103987. <https://doi.org/10.1016/j.compind.2023.103987>.
- Phua, A., Davies, C.H.J., Delaney, G.W., 2022. A digital twin hierarchy for metal additive manufacturing. *Comput. Ind.* 140, 103667. <https://doi.org/10.1016/j.compind.2022.103667>.
- Pires, F., Leitão, P., Moreira, A.P., Ahmad, B., 2023. Reinforcement learning based trustworthy recommendation model for digital twin-driven decision-support in manufacturing systems. *Comput. Ind.* 148 (February), 0–2. <https://doi.org/10.1016/j.compind.2023.103884>.
- Plavšić, J., Mišković, I., 2024. VR-based digital twin for remote monitoring of mining equipment: architecture and a case study. *Virtual Real. Intell. Hardw.* 6 (2), 100–112. <https://doi.org/10.1016/j.vrih.2023.12.002>.
- Powell, D., Magnanini, M.C., Colledani, M., Myklebust, O., 2022. Advancing zero defect manufacturing: a state-of-the-art perspective and future research directions. *Comput. Ind.* 136, 103596. <https://doi.org/10.1016/j.compind.2021.103596>.
- Priyanka, E.B., Thangavel, S., Gao, X.Z., Sivakumar, N.S., 2022. Digital twin for oil pipeline risk estimation using prognostic and machine learning techniques. *J. Ind. Inf. Integr.* 26, 100272. <https://doi.org/10.1016/j.jii.2021.100272>.
- Psarommatas, F., Fraile, F., Ameri, F., 2023. Zero defect manufacturing ontology: a preliminary version based on standardised terms. *Comput. Ind.* 145 (December), 103832. <https://doi.org/10.1016/j.compind.2022.103832>.
- Rahman, M., Liu, H., Masri, M., Durazo-Cardenas, I., Starr, A., 2023. A railway track reconstruction method using robotic vision on a mobile manipulator: a proposed strategy. *Comput. Ind.* 148, 103900. <https://doi.org/10.1016/j.compind.2023.103900>.
- Reifsnider, K., Majumdar, P., 2013. Multiphysics stimulated simulation digital twin methods for fleet management. *Collect. Tech. Pap. AIAA/ASME/ASCE/AHS/ASC Struct. Struct. Dyn. Mater. Conf.* 1–11.
- Rico, M., Taverna, M.L., Galli, M.R., Calusco, M.L., 2023. Context-aware representation of digital twins' data: the ontology network role. *Comput. Ind.* 146 (January). <https://doi.org/10.1016/j.compind.2023.103856>.
- Robles, J., Martín, C., Díaz, M., 2023. OpenTwins: an open-source framework for the development of next-gen compositional digital twins. *Comput. Ind.* 152 (May), 104007. <https://doi.org/10.1016/j.compind.2023.104007>.
- Rocca, R., Rosa, P., Sassanelli, C., Fumagalli, L., Terzi, S., 2020. Integrating virtual reality and digital twin in circular economy practices: a laboratory application case. *Sustainability* 12 (6). <https://doi.org/10.3390/su12062286>.
- Rodríguez, F., Chicaiza, W.D., Sánchez, A., Escano, J.M., 2023. Updating digital twins: methodology for data accuracy quality control using machine learning techniques. *Comput. Ind.* 151 (April), 103958. <https://doi.org/10.1016/j.compind.2023.103958>.
- Rongen, S., Nikolova, N., van der Pas, M., 2023. Modelling with AAS and RDF in industry 4.0. *Comput. Ind.* 148 (957204), 103910. <https://doi.org/10.1016/j.compind.2023.103910>.
- Rouast, P.V., Adam, M.T.P., Chiong, R., 2021. Deep learning for human affect recognition: insights and new developments. *IEEE Trans. Affect. Comput.* 12 (2), 524–543. <https://doi.org/10.1109/TAFFC.2018.2890471>.
- Rowan, N.J., 2024. Digital technologies to unlock safe and sustainable opportunities for medical device and healthcare sectors with a focus on the combined use of digital twin and extended reality applications: a review. *Sci. Total Environ.* 926 (February), 171672. <https://doi.org/10.1016/j.scitotenv.2024.171672>.
- Ruohomäki, T., Airaksinen, E., Huuska, P., Kesäniemi, O., Martikka, M., Suomisto, J., 2018. Smart city platform enabling digital twin. 2018 Int. Conf. Intell. Syst. (IS) 0, 155–161. <https://doi.org/10.1109/IS.2018.8710517>.
- Saeed, A., Alam, A.K., Azam, A., Khalid, M., Tauni, O., 2022. Game-Based interventions as support for learning difficulties and knowledge enhancement in patients with dyslexia: a systematic literature review. *EAI/Springer Innov. Commun. Comput.* https://doi.org/10.1007/978-3-030-75123-4_4.
- Saini, G., Ashok, P., van Oort, E., Isbell, M.R., 2018. Accelerating well construction using a digital twin demonstrated on unconventional well data in North America. *SPE/AAPG/SEG Unconv. Resour. Technol. Conf.* 2018 URTC 2018 113. <https://doi.org/10.15530/urtec-2018-2902186>.
- Scurati, G.W., Ferrise, F., Bertoni, M., 2025. Extended realities and discrete events simulations: a systematic review to define design trade-offs and directions. *Comput. Ind.* 164 (August), 104188. <https://doi.org/10.1016/j.compind.2024.104188>.
- Seidel, R., Rachinger, B., Thielen, N., Schmidt, K., Meier, S., Franke, J., 2023. Development and validation of a digital twin framework for SMT manufacturing. *Comput. Ind.* 145 (May). <https://doi.org/10.1016/j.compind.2022.103831>.

- Semeraro, C., Lezoche, M., Panetto, H., Dassisti, M., 2021. Digital twin paradigm: a systematic literature review. *Comput. Ind.* 130. <https://doi.org/10.1016/j.compind.2021.103469>.
- Sengupta, S., Dreyer, H., 2023. Realizing zero-waste value chains through digital twin-driven S&OP: a case of grocery retail. *Comput. Ind.* 148 (September), 103890. <https://doi.org/10.1016/j.compind.2023.103890>.
- Sharma, A., Kosasih, E., Zhang, J., Brintrup, A., Calinescu, A., 2022. Digital twins: state of the art theory and practice, challenges, and open research questions. *J. Ind. Inf. Integr.* 30 (November), 100383. <https://doi.org/10.1016/j.jii.2022.100383>.
- Sierla, S., Kyrki, V., Aarnio, P., Vyatkin, V., 2018. Automatic assembly planning based on digital product descriptions. *Comput. Ind.* 97, 34–46. <https://doi.org/10.1016/j.compind.2018.01.013>.
- Singh, K., Goyal, V., Rana, P., 2021. Existing assistive techniques for dyslexics: a systematic review. *Artif. Intell. Accurate Anal. Detect. Autism Spectr. Disord.* 94–104.
- Singh, M., Srivastava, R., Fuenmayor, E., Kuts, V., Qiao, Y., Murray, N., Devine, D., 2022. Applications of digital twin across industries: a review (MDPI). *Appl. Sci.* 12 (11). <https://doi.org/10.3390/app12115727>.
- Stegmaier, V., Jazdi, N., Weyrich, M., 2024. A method for the automated digitalisation of fluid circuit diagrams. *Comput. Ind.* 162 (July), 104139. <https://doi.org/10.1016/j.compind.2024.104139>.
- Su, C., Han, Y., Tang, X., Jiang, Q., Wang, T., He, Q., 2024. Knowledge-based digital twin system: using a knowledge-driven approach for manufacturing process modeling. 159–160(February). *Comput. Ind.*, 104101. <https://doi.org/10.1016/j.compind.2024.104101>.
- Suhail, S., Iqbal, M., Hussain, R., Jurdak, R., 2023. ENIGMA: an explainable digital twin security solution for cyber-physical systems. *Comput. Ind.* 151 (May), 103961. <https://doi.org/10.1016/j.compind.2023.103961>.
- Suhail, S., Malik, S.U.R., Jurdak, R., Hussain, R., Matulevičius, R., Svetinovic, D., 2022. Towards situational aware cyber-physical systems: a security-enhancing use case of blockchain-based digital twins. *Comput. Ind.* 141. <https://doi.org/10.1016/j.compind.2022.103699>.
- Tao, F., Xiao, B., Qi, Q., Cheng, J., Ji, P., 2022. Digital twin modeling. *J. Manuf. Syst.* 64, 372–389. <https://doi.org/10.1016/j.jmsy.2022.06.015>.
- Tapia, E., Lopez-Novoa, U., Sastoque-Pinilla, L., López-de-Lacalle, L.N., 2024. Implementation of a scalable platform for real-time monitoring of machine tools. *Comput. Ind.* 155 (October), 104065. <https://doi.org/10.1016/j.compind.2023.104065>.
- Toothman, M., Braun, B., Bury, S.J., Moyne, J., Tilbury, D.M., Ye, Y., Barton, K., 2023. A digital twin framework for prognostics and health management. *Comput. Ind.* 150 (May), 103948. <https://doi.org/10.1016/j.compind.2023.103948>.
- Tuegel, E.J., Ingrassia, A.R., Eason, T.G., Spottswood, S.M., 2011. Reengineering aircraft structural life prediction using a digital twin. *Int. J. Aerosp. Eng.*, 154798 <https://doi.org/10.1155/2011/154798>.
- Valckenaers, P., 2020. Perspective on holonic manufacturing systems: PROSA becomes ARTI. *Comput. Ind.* 120, 103226. <https://doi.org/10.1016/j.compind.2020.103226>.
- Verdouw, C., Tekinerdogan, B., Beulens, A., Wolfert, S., 2021. Digital twins in smart farming. *Agric. Syst.* 189 (January), 103046. <https://doi.org/10.1016/j.agry.2020.103046>.
- Wang, T., Tan, C., Huang, L., Shi, Y., Yue, T., Huang, Z., 2023. Simplicity tested: a model-based digital twin tested. *Comput. Ind.* 145 (April), 103804. <https://doi.org/10.1016/j.compind.2022.103804>.
- Wang, X.V., Wang, L., 2019. Digital twin-based WEEE recycling, recovery and remanufacturing in the background of industry 4.0. *Int. J. Prod. Res.* 57 (12), 3892–3902. <https://doi.org/10.1080/00207543.2018.1497819>.
- Wang, X., Yu, H., McGee, W., Menassa, C.C., Kamat, V.R., 2024a. Enabling building information Model-driven human-robot collaborative construction workflows with closed-loop digital twins. *Comput. Ind.* 161 (December), 104112. <https://doi.org/10.1016/j.compind.2024.104112>.
- Wang, S., Zhang, J., Wang, P., Law, J., Calinescu, R., Mihaylova, L., 2024b. A deep learning-enhanced digital twin framework for improving safety and reliability in human-robot collaborative manufacturing. *Robot. Comput. Integr. Manuf.* 85. <https://doi.org/10.1016/j.rcim.2023.102608>.
- Wang, B., Zhou, H., Li, X., Yang, G., Zheng, P., Song, C., Yuan, Y., Wuest, T., Yang, H., Wang, L., 2024c. Human digital twin in the context of industry 5.0. In: *In Robotics and Computer-Integrated Manufacturing*. Elsevier Ltd. <https://doi.org/10.1016/j.rcim.2023.102626>.
- Winning, K., Solutions, P.D., 2018. Lesson learned: creating the digital twin concurrently for pipeline projects. *Esri Eur. Pet. GIS Conf.* 1–23.
- Woitsch, R., Sumereder, A., Falcioni, D., 2022. Model-based data integration along the product & service life cycle supported by digital twinning. *Comput. Ind.* 140, 103648. <https://doi.org/10.1016/j.compind.2022.103648>.
- Xia, D., Shi, J., Wan, K., Wan, J., Martinez-Garcia, M., Guan, X., 2023a. Digital twin and artificial intelligence for intelligent planning and Energy-Efficient deployment of 6G networks in smart factories. *IEEE Wirel. Commun.* 30 (3), 171–179. <https://doi.org/10.1109/MWC.017.2200495>.
- Xia, P., Xu, F., Song, Z., Li, S., Du, J., 2023b. Sensory augmentation for subsea robot teleoperation. *Comput. Ind.* 145 (June), 103836. <https://doi.org/10.1016/j.compind.2022.103836>.
- Xu, Y., Qamsane, Y., Puchala, S., Januszczak, A., Tilbury, D.M., Barton, K., 2024a. A data-driven approach toward a machine- and system-level performance monitoring digital twin for production lines, 157–158(February). *Comput. Ind.*, 104086. <https://doi.org/10.1016/j.compind.2024.104086>.
- Xu, Z., Wang, Z., Gao, C., Zhang, K., Lv, J., Wang, J., Liu, L., 2024b. A digital twin system for centrifugal pump fault diagnosis driven by transfer learning based on graph convolutional neural networks. *Comput. Ind.* 163 (August), 104155. <https://doi.org/10.1016/j.compind.2024.104155>.
- Yang, C., Cai, B., Zhang, R., Zou, Z., Kong, X., Shao, X., Liu, Y., Shao, H., Akbar Khan, J., 2023. Cross-validation enhanced digital twin driven fault diagnosis methodology for minor faults of subsea production control system. *Mech. Syst. Signal Process.* 204. <https://doi.org/10.1016/j.ymssp.2023.110813>.
- Ye, Z., Ye, Y., Zhang, C., Zhang, Z., Li, W., Wang, X., Wang, L., Wang, L., 2023. A digital twin approach for tunnel construction safety early warning and management. *Comput. Ind.* 144 (April), 103783. <https://doi.org/10.1016/j.compind.2022.103783>.
- Zeb, A., Kortelainen, J., Rantala, T., Saunila, M., Ukko, J., 2022. On the alleviation of imminent technical and business challenges of long-lasting functional digital twins. *Comput. Ind.* 141, 103701. <https://doi.org/10.1016/j.compind.2022.103701>.
- Zhang, J., Luo, H., Xu, J., 2022. Towards fully BIM-enabled building automation and robotics: a perspective of lifecycle information flow. *Comput. Ind.* 135, 103570. <https://doi.org/10.1016/j.compind.2021.103570>.
- Zhang, X., Wang, X., Yao, H., Luo, W., Qiang, Z., Luo, D., 2025. Multi-similarity and gradient fusion digital twins for fault detection and diagnosis of rolling bearings. *Comput. Ind.* 168, 104273. <https://doi.org/10.1016/j.compind.2025.104273>.



Peipei Ding is currently a Ph.D. student in the Discipline of Management Science and Engineering at the School of Economics and Management, Fuzhou University, China. She won the first-prize PhD Fellowship of Fuzhou University in 2023. Her research interests include smart logistics, mining optimisation and inventory management.



Shi Qiang Liu is currently a full professor with the School of Economics and Management at Fuzhou University. He received his Bachelor's and Master's degrees from Harbin Institute of Technology (HIT), another Master's degree in industrial and systems engineering from the National University of Singapore (NUS), and a Ph.D. degree in operations research from the School of Mathematical Sciences, Queensland University of Technology (QUT) in Australia. He worked in Australia for over 12 years, as a Research Scientist in CRC ORE and as a Research Fellow at QUT. He has solid background and expertise in operations research, machine learning, scheduling theory and applications in many industries such as mining, railway, robotics, manufacturing, and healthcare.



funding.

Raymond Chiong received the Ph.D. degree in computer science from the University of Melbourne, Parkville, VIC, Australia. He is currently a professor with the School of Science and Technology, University of New England, Parramatta, NSW, Australia. He is also an honorary professor with the School of Medicine and Public Health, University of Newcastle, Callaghan, NSW, Australia. He has been actively pursuing research related to the use of automated intelligent computing methods, including machine learning and optimisation algorithms, with applications in energy load forecasting, energy-efficient production scheduling, and computer vision. To date, he has produced/co-authored more than 260 refereed publications and attracted over 6 million dollars in research and industry



Sandeep Dhakal received the M.Sc. degree in artificial intelligence from the University of St Andrews, St Andrews, U.K., in 2011. He received his Ph.D. from the University of Newcastle, Australia, in 2023. He is currently a casual academic staff member with the University of New England, Australia. His research interests include digital twins, agent-based modelling, evolutionary game theory, and machine learning.



Dewang Chen received the Ph.D. degree in control theory and control engineering from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2003. From 2011–2015, he was a full professor with the State Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, Beijing. In 2009, he was a visiting scholar with the Department of Electrical Engineering and Computer Science, College of Engineering, University of California at Berkeley, Berkeley, CA, USA. He is currently a Minjiang Distinguished Professor with the School of Transportation, Fujian University of Technology, Fuzhou, China. His current research interests include artificial intelligence, soft computing, and intelligent transportation systems. Dr. Chen has served on the programme committees for more than 15 international conferences and as an associate editor for the *IEEE Transactions on Intelligent Transportation Systems*.



Debiao Li received the Ph.D. degree in system science from The State University of New York at Binghamton, Binghamton, NY, USA, in 2015. He is currently a professor with the School of Economics and Management, Fuzhou University, China. His current research interests include scheduling, operations management, and artificial intelligence.



Hoi-Lam Ma is an Associate Head and Associate Professor in Supply Chain and Information Management at Hang Seng University of Hong Kong. She received her Ph.D. degree from the Hong Kong Polytechnic University. Her research interests are in airline operations, container terminal operations, logistics and supply chain management, and scheduling optimisation. She has published over 30 journal papers. She serves as an editorial board member for *Transportation Research Part E*, *Industrial Management & Data Systems*, and the *Journal of Air Transport Management*. Additionally, she has edited several special issues in reputable journals.



Sai-Ho Chung received the Ph.D. degree in industrial and manufacturing systems engineering from the University of Hong Kong in 2007. He is currently an associate professor with the Department of Industrial and Systems Engineering, Hong Kong Polytechnic University, Hong Kong. He has authored or co-authored more than 100 journal papers. His research interests include logistics and supply chain management, supply chain finance, production scheduling, distribution networks, machine learning, container port terminals, and aviation. He serves as an editorial board member for *Transportation Research Part E* and has edited several special issues in reputable journals.